

Is Liquidity Risk Priced in Partially Segmented Markets?*

Ines Chaieb Vihang Errunza Hugues Langlois
University of Geneva and McGill University HEC Paris
Swiss Finance Institute

November 22, 2016

Abstract

We analyze the impact of liquidity costs and market segmentation on asset pricing. The freely traded securities command a global market risk premium and world liquidity risk premia whereas the securities that can be held by only a subset of investors command additionally a conditional local market risk premium and conditional local liquidity risk premia. Our model provides a formal framework for testing important issues such as liquidity levels and risks in a realistic world market setting and for examining the interaction between investability and illiquidity. Empirical test results for a sample of 21 emerging markets strongly support theoretical predictions.

JEL Classification: G12, G15, F30, G20, G30

Keywords: International Asset Pricing, Liquidity Risk, Transaction Cost, Emerging Markets, Market Integration.

*We are thankful for helpful comments from Francesca Carrieri, Thierry Foucault, Jiro Kondo, Kuan-Hui Lee, Ioanid Rosu, Sergei Sarkissian, and seminar participants at McGill University. Chaieb is at University of Geneva and Swiss Finance Institute, Geneva, Switzerland, email: ines.chaieb@unige.ch. Errunza is at McGill University, Desautels Faculty of Management, Canada, email: vihang.errunza@mcgill.ca and Langlois is at HEC Paris, France, email: langlois@hec.fr. We acknowledge excellent research assistance from Devinder Paul Singh. Chaieb acknowledges financial support from the Geneva Finance Research Institute and the Swiss Finance Institute research grant 2012-1-006, Errunza acknowledges financial support from the Bank of Montreal Chair at McGill University, IFM2 and SSHRC. Langlois is grateful for financial support from the Investissements d'Avenir Labex (ANR-11-IDEX-0003/Labex Ecodec/ANR-11-LABX-0047).

1 Introduction

Liquidity costs and their uncertain variation affect the willingness and the ability of foreign investors to invest in emerging markets (EMs). Unfortunately, most of the asset pricing models under barriers to portfolio flows do not take into account liquidity concerns¹. We analyze the effect of illiquidity cost and systematic liquidity risk factors on the pricing of EM securities by developing a formal international asset pricing model (IAPM) that takes into account cross-border investment barriers as well as random transaction costs. Our pricing equation under partial segmentation allows us to test important issues such as liquidity levels and risks in a realistic world market setting and to examine the interaction between investability and illiquidity. Although our empirical set up focusses on EMs, it is applicable to many global markets characterized by explicit or implicit barriers to free cross-border portfolio flows.²

Our new liquidity IAPM exploits the theoretical insights of international frameworks under partial segmentation. In a two country framework, we assume that all domestic securities are investable and hence can be freely traded by all investors whereas the foreign market consists of two segments: (a) a non-investable set of securities that can be held only by local investors³, and (b) an investable set that can be held by all investors. We include random transaction costs on both investable and non-investable securities⁴. Although the investable securities are priced as in [Acharya and Pedersen \(2005\)](#) [AP hereafter], the market risk and liquidity shocks are aggregated at the world level. Except for the Jensen's inequality term, our pricing equation for investables can be viewed as an extension of AP to an international setup. Additionally, non-investable securities command a conditional market risk premium and conditional liquidity risk premiums. The conditional market risk is a residual risk that stems from imperfect spanning of the returns on the non-investable segment of the foreign market by returns on substitute assets.⁵ All of the sensitivity of returns to aggregate local market risk, sensitivity of returns to aggregate local market liquidity, the sensitivity

¹See for example, [Stulz \(1981\)](#), [Errunza and Losq \(1985\)](#) [hereafter EL], [Eun and Janakiraman \(1986\)](#), [de Jong and Roon \(2005\)](#), [Chaieb and Errunza \(2007\)](#).

²Explicit barriers include legal restrictions on ownership, foreign exchange transactions, repatriation of profits, whereas implicit barriers encompass institutional, informational, governance, and market development variables.

³Non-investable assets in the foreign market cannot be or are not traded by domestic investors due to severe explicit and/or implicit barriers.

⁴Liquidity could itself be endogenously determined by other implicit barriers such as asymmetric information (see [Johnson, 2006](#); [Huang and Wang, 2009](#), among others).

⁵Substitute assets consist of all assets that can be freely traded by all investors including the world market portfolios, closed-end country funds, exchange traded funds, ADRs, and GDRs.

of liquidity to aggregate local market risk, and the local commonality in liquidity are conditional on returns on the substitute assets net of the illiquidity costs. The prices of global market risk and of global liquidity risks are the same. Similarly, the prices of local market risk and local liquidity risks are the same. Thus, our model extends the well-established theoretical asset pricing literature regarding the importance of liquidity for U.S. and other integrated developed markets (see, for example, [Amihud and Mendelson, 1986](#); [Pastor and Stambaugh, 2003](#); [Acharya and Pedersen, 2005](#)). Indeed, our model nests the AP model in the absence of investability constraints, the EL model in the absence of liquidity and the [Grauer, Litzenberger, and Stehle \(1976\)](#) perfect world market model in the absence of both the investability constraints and liquidity.

Our paper is close to the work of [Lee \(2011\)](#) and [Bekaert, Harvey, and Lundblad \(2007\)](#) [BHL hereafter]. Whereas Lee adopts an econometric specification to obtain his testable equations, BHL analytically derive stylized pricing equations under market integration and segmentation. Furthermore BHL assume that liquidity is a second risk factor that affects the world pricing kernel in addition to the world market risk factor and hence triggers its own price of liquidity risk. Our work also adds to the vast empirical literature on the relation between the average stock returns and liquidity documented in several U.S. studies (see, for example, [Brennan, Chordia, and Subrahmanyam, 1998](#)), commonality in liquidity (see, for example, [Chordia, Roll, and Subrahmanyam, 2000](#); [Hasbrouck and Seppi, 2001](#); [Huberman and Halka, 2001](#); [Koch, Ruenzi, and Starks, 2014](#), for evidence from the U.S.)⁶, as well as a number of empirical papers that investigate liquidity effects in international markets, (see, for example, [Domowitz, Glen, and Madhavan, 2001](#); [Lesmond, 2005](#); [Bekaert et al., 2007](#); [Lee, 2011](#); [Goyenko and Sarkissian, 2014](#); [Karolyi, Lee, and van Dijk, 2012](#); [Amihud, Hameed, Kang, and Zhang, 2015](#))⁷. The above studies underscore the importance

⁶Additionally, there is ample evidence that liquidity risk is unconditionally priced for the U.S. market although the contribution of the liquidity risk premium varies across studies and depends on the proxy used for liquidity, see for example, [Pastor and Stambaugh \(2003\)](#), [Acharya and Pedersen \(2005\)](#), [Sadka \(2006\)](#), [Korajczyk and Sadka \(2008\)](#), and [Kim and Lee \(2014\)](#). [Watanabe and Watanabe \(2008\)](#) investigate the dynamic nature of illiquidity premium and show that liquidity betas and liquidity risk premiums are larger when investors face a higher level of preference uncertainty. [Hagstromer, Hansson, and Nilsson \(2011\)](#) test a conditional version of the AP liquidity-adjusted CAPM and find significant time-variation in the illiquidity premium but commonality risk is the least important component.

⁷Specifically, [Domowitz et al. \(2001\)](#) document that transaction costs in emerging markets are significantly higher than in developed markets. [Lesmond \(2005\)](#) shows that liquidity costs are higher in countries with weak legal enforcement. BHL find that illiquidity predicts returns in emerging markets and that the level of liquidity is more important than systematic liquidity risk premiums. Based on an econometric model in an unconditional set up, [Lee \(2011\)](#) finds that liquidity risk is priced. [Goyenko and Sarkissian \(2014\)](#) analyze the impact of changes in the liquidity of US Treasury bills on global equity returns. [Karolyi et al. \(2012\)](#) show that commonality in liquidity is greater for emerging markets than for developed markets though it trended down for many countries. They also find that commonality in liquidity increases with capital inflows, while it decreases with market openness. [Amihud](#)

of liquidity in the U.S. and international markets. Our focus on pricing liquidity risk for investables and non-investables within the context of a formal asset pricing model that accounts for the time-varying partially segmented nature of EMs, along with the flexible empirical methodology used in the estimation, sets the paper apart from the earlier literature.

To test our asset pricing model, we need to define the investable and non-investable set of securities. We follow [Karolyi and Wu \(2012\)](#) and use cross-listings to characterize investable stocks⁸. We also need a measure of illiquidity costs. In the theoretical set up, the exogenous illiquidity cost of a security is modeled as the per trade cost. The illiquidity cost could then be measured with the bid-ask spread. But the data are available only for short sample periods and for a small set of emerging countries limiting their use for validation of the asset pricing model and for studying the effect of illiquidity on expected returns. Alternatively, we could use the [Amihud \(2002\)](#) absolute return to dollar volume ratio which is related to measures of price impact and fixed trading costs or the incidence of zero returns of [Lesmond, Ogden, and Trzcinka \(1999\)](#) which could proxy for search costs.⁹ The applicability of the price impact measure is also limited by the scarcity of volume data in EMs. Moreover, as we require a measure of the cost of trade to test our asset pricing model, we need to scale these two proxies as in, for example, AP. The paucity of effective bid-ask spread data for emerging markets limits such computation.¹⁰ Hence, we use stock-level daily high and low prices, for which long sample periods are available, to construct the bid-ask spread proxy of [Corwin and Schultz \(2012\)](#). They use the high-low price range for two consecutive days and their two-day high-low price range to disentangle volatility from the bid-ask spread.

We first estimate diversification portfolios (DPs) which are the dynamic replicating portfolios

et al. (2015) provide international evidence on the positive relation between expected return and illiquidity, using data from 45 countries including 19 emerging markets. See also [Amihud and Mendelson \(2015\)](#) for a review of the work on the effect of liquidity on expected returns.

⁸Although BHL do not formally incorporate the impact of changing market structure on pricing of liquidity risk, they do suggest that expected returns should decline post market liberalization which is related to investability.

⁹[Lesmond et al. \(1999\)](#) show that the proportion of zero returns measure is highly correlated with the effective spread. This measure has been applied in a number of recent papers. See for example, BHL, [Lesmond, Schill, and Zhou \(2004\)](#), [Lang, Lins, and Maffett \(2012\)](#), and [Mei, Scheinkman, and Xiong \(2009\)](#) among others. [Levine and Schmukler \(2006\)](#) use both the proportion of zero returns and the Amihud's measure to examine the relationship between home market liquidity and cross listed trading across 31 home countries. [Amihud \(2002\)](#)'s widely applied price impact measure has been recently used in [Karolyi et al. \(2012\)](#) to analyze the commonality patterns of liquidity, returns, and turnover across 40 countries. [Korajczyk and Sadka \(2008\)](#), [Mancini, Rinaldo, and Wrampelmeyer \(2013\)](#) and [Kim and Lee \(2014\)](#) use a latent liquidity measure constructed from a principal component across different liquidity measures. See also [Fong, Holden, and Trzcinka \(2014\)](#) for a recent review of the different liquidity proxies used in international studies.

¹⁰See BHL for a detailed discussion of trading costs data issues in EMs.

obtained by projecting the non-investables returns on substitute assets returns. Our methodology has two important features. First, the weights of the DP are time-varying and computed from the conditional covariance matrices. The dynamics of these conditional covariances are the same as the ones used in the IAPM, and hence the construction of the DP is consistent with the asset pricing framework. Second, our methodology can account for the variation over time in the set of substitute assets as cross-listings, country funds, and ETFs are issued or delisted. To estimate the conditional covariance matrices we proceed in two steps. To estimate variances we first fit an AR-NGARCH on net of transaction cost returns for non-investable securities and all substitute assets. In a second step, we estimate a dynamic copula by using all available assets at each point in time as in [Christoffersen, Jacobs, Jin, and Langlois \(2016\)](#). We then use the dynamic variances from the first step estimations and correlations from the second step estimation to compute the DP weights.¹¹

Next, we run asset pricing tests in a conditional set up of our liquidity-adjusted IAPM. First, we find significant and time-varying prices of world covariance risk. Its magnitude of between one and two is very plausible for the world aggregate risk aversion. Second, we find significant and time-varying prices of local risk. The cross-sectional average of the prices of local covariance risk is 3.6 with significant variation across markets. The price of local risk is proportional to the differential in risk aversion between an EM average investor and the world average investor and hence again seems economically plausible. Third, we obtain significant reductions in pricing errors when we include illiquidity costs in the asset pricing model. Fourth, we find that the total contribution of illiquidity level and risk to risk premiums is on average 0.75% annually across EMs and its magnitude is time varying and higher in crisis periods. Our results are robust to including all EM stocks in the non-investable market portfolio.

The article is organized as follows. Section 2 models the effect of liquidity on expected returns in partially segmented markets. Section 3 outlines the empirical methodology. Section 4 presents data and summary statistics. Section 5 reports the results, and examines the statistical and economic

¹¹The past literature has used step wise regressions with forward and backward threshold criteria that preserves assets with the most significant coefficients to estimate DP while allowing for some dynamics in the weights through dummy variables set to one upon the availability of new cross-listings. Our new approach of constructing the DPs with dynamic weights computed from time-varying second moments is internally consistent because the same second moment dynamics are used for asset pricing tests. Also our approach ensures that the different parametrization of the moments conditional on the set of substitute assets are equivalent which is not the case with the approach to construct DP used in past literature.

significance of the liquidity risk premiums. Section 6 presents a robustness check using alternative specifications of the asset pricing equations and of the non-investable set. Section 7 concludes.

2 The asset pricing model

We postulate a two country global capital market: the domestic country D and the foreign country F . We label all securities that can be freely traded by all investors as investable and those that can be held by only foreign investors as non-investable. We can view the domestic market as a well-developed market (such as the United States) that is open to all investors and the foreign market as an EM where foreign participation is limited due to explicit and/or implicit barriers. In this setting, domestic (U.S.) investors can hold domestic securities and the investable segment of the foreign market whereas foreign (EM) investors can freely trade in the domestic and the foreign market. This characterization of the global market is fairly realistic. Of course, a market structure, in which both countries face some segmentation, is more attractive. However, our simplification makes understanding the forces at work easier without much loss of generality.¹²

2.1 Assumptions and notations

We use the subscript i and n as a generic index to represent the investable and non-investable securities respectively, bold letters to denote vectors, and capital letters to denote matrices or portfolio level variables.

- **A1: Market Structure.** There are N risky assets partitioned as follow: The first N_I securities are investable securities and the last N_N assets are non-investable securities. The vector of log-returns can be partitioned as $\mathbf{r}_t \equiv \begin{pmatrix} \mathbf{r}_{i,t} \\ \mathbf{r}_{n,t} \end{pmatrix}$. The vector of market values at time t , $\mathbf{M}_t$, is similarly partitioned as $\mathbf{M}_t \equiv \begin{pmatrix} \mathbf{M}_{i,t} \\ \mathbf{M}_{n,t} \end{pmatrix}$. $M_{N,t}$ is the market capitalization of non-investables at time t defined as $M_{N,t} = \sum_{n=1}^{N_N} M_{n,t}$. Agents can also borrow and lend at the domestic risk-free log-rate of return r_f , which is exogenous.

¹²Indeed, as shown by Chaieb and Errunza (2007), the results carry through to the more general market structure with the addition of another premium at equilibrium.

- **A2: Agents.** Every period, the economy has two agents with CRRA preferences. The first agent is domestic and is restricted to investing only in investable securities. The second agent is foreign and can invest in both non-investable and investable securities. Agents live for one period, invest at time t , and consume at $t+1$. Hence both investors have the same one-period trading horizon and holding period i.e. the trading frequency and holding period are set exogenously. This is a strong assumption. Allowing for endogenous trading frequency will result in less trading for illiquid assets. [Constantinides \(1986\)](#) shows that transaction costs might then have second-order effects on liquidity premia.
- **A3: Transaction costs.** Transaction costs denoted by the N -by-1 vector C_t are random and include bid-ask spread as well as search costs. They are modeled as the proportional cost of buying and selling an investable or non-investable security j . Short-selling is allowed. Both long and short holders pay transaction costs. As in AP, the one-period investors investing at $t-1$ pay transaction costs proportional to the current price when closing the position at time t . The net log-returns of the agents can then be written as $r_{j,t}^{net} = r_{j,t} - c_{j,t}$ for the investors who are long in asset j and as $r_{j,t}^{net} = -r_{j,t} - c_{j,t}$ for investors who are short in asset j , where $c_{j,t} = -\ln(1 - C_{j,t})$ (see [Bongaerts, Jong, and Driessen, 2011](#), for a similar assumption). We assume that the vector of net log-returns $\mathbf{r}_t^{net}$ is normally distributed and denote Σ_t as its N -by- N conditional variance-covariance matrix which can be partitioned as $\Sigma_t \equiv \begin{pmatrix} \Sigma_{ii,t} & \Sigma_{in,t} \\ \Sigma_{in,t}^\top & \Sigma_{nn,t} \end{pmatrix}$, where $\Sigma_{ii,t}$ is the variance-covariance matrix of the investable assets, $\Sigma_{nn,t}$ is the variance-covariance matrix of the non-investable securities, and $\Sigma_{in,t}$ is the covariance matrix between investable and non-investable securities.
- **A4:** The returns are measured in domestic currency, the reference currency. We follow the asset pricing literature on barriers to cross-border investments and assume that the purchasing power parity holds (see, for example, [Stulz, 1981](#)).

2.2 Asset demands

The domestic investor d can invest in the riskless bond and the N_I investable risky assets. Let the vector $\omega_{d,t}$ denote the fraction of his wealth invested in the risky assets at time t and γ^d his coefficient of relative risk aversion. Investor d maximizes at time t the utility of his terminal wealth

W_{t+1}^d , that is,

$$\max_{\boldsymbol{\omega}_{d,t}} E_t \left[\left(\frac{W_{t+1}^d}{1 - \gamma^d} \right)^{1 - \gamma^d} \right]$$

subject to the budget constraint $W_{t+1}^d = (1 + R_{P,t+1}^d) W_t^d$ where $R_{P,t+1}^d = \boldsymbol{\omega}_{d,t}^\top (e^{\mathbf{r}_{i,t+1}^{net}} - e^{r_f} \boldsymbol{\iota}_{N_I}) + e^{r_f}$ is the simple net return on the portfolio of investor d and $\boldsymbol{\iota}_{N_I}$ is an $(N_I \times 1)$ vector of ones. Assuming the portfolio return is log-normal and using the [Campbell and Viceira \(2002\)](#) discrete-time approximation to relate asset log-returns to portfolio log-returns¹³, the optimization can be restated as,

$$\max_{\boldsymbol{\omega}_{d,t}} \boldsymbol{\omega}_{d,t}^\top E_t \left[\mathbf{r}_{i,t+1} - \mathbf{c}_{i,t+1} - r_f \boldsymbol{\iota}_{N_I} + \frac{1}{2} \boldsymbol{\sigma}_{i,t+1}^2 \right] - \frac{1}{2} \gamma^d \boldsymbol{\omega}_{d,t}^\top \Sigma_{ii,t+1} \boldsymbol{\omega}_{d,t}$$

where $\boldsymbol{\sigma}_{i,t+1}^2$ is the vector containing the variance of investable asset net log-returns i.e. the diagonal elements of $\Sigma_{ii,t+1}$. From first order conditions, the solution for the vector of optimal portfolio weights is,

$$\boldsymbol{\omega}_{d,t} = \frac{1}{\gamma^d} \Sigma_{ii,t+1}^{-1} E_t \left[\mathbf{r}_{i,t+1} - \mathbf{c}_{i,t+1} - r_f \boldsymbol{\iota}_{N_I} + \frac{1}{2} \boldsymbol{\sigma}_{i,t+1}^2 \right].$$

A foreign investor f can invest in the riskless bond, the domestic securities and the foreign securities. Given his investment opportunity set, he solves a similar optimization problem as the domestic investor. The vector $\boldsymbol{\omega}_{f,t}$ of optimal portfolio weights for the foreign unrestricted investor can be written as,

$$\boldsymbol{\omega}_{f,t} = \frac{1}{\gamma^f} \Sigma_{t+1}^{-1} E_t \left[\mathbf{r}_{t+1} - \mathbf{c}_{t+1} - r_f \boldsymbol{\iota}_N + \frac{1}{2} \boldsymbol{\sigma}_{t+1}^2 \right]$$

where $\boldsymbol{\sigma}_{t+1}^2$ is the vector containing the variance of investable and non-investable asset net log-returns i.e. the diagonal elements of Σ_{t+1} , γ^f is the coefficient of relative risk aversion of the foreign investor, and $\boldsymbol{\iota}_N$ is an $(N \times 1)$ vector of ones.

Using market clearing conditions and assuming that equities are in net positive supply equal to their market capitalization and the bonds are in zero net supply, we obtain the equilibrium asset pricing relationships.

¹³[Campbell and Viceira \(2002\)](#) use a Taylor approximation of the non-linear function relating log asset returns to log portfolio returns.

2.3 Equilibrium risk and return

2.3.1 Investable set

The investable securities are priced as if the market is fully integrated. The expected net excess return for an investable security $i \in \{1, \dots, N_I\}$ is given by,

$$E_t [r_{i,t+1}^{net} - r_f] + \frac{1}{2} \text{var}_t (r_{i,t+1}^{net}) = \gamma_t \text{cov}_t (r_{i,t+1}^{net}, r_{W,t+1}^{net}) \quad (1)$$

where $\gamma_t = \left(\frac{W_t^d}{\gamma^d} + \frac{W_t^f}{\gamma^f} \right)^{-1}$ W_t is the aggregate relative risk aversion coefficient, $W \equiv W^d + W^f$ is the aggregate global wealth and $r_{W,t+1}^{net}$ is the net log-return on the global market portfolio. We then express the model in terms of gross returns,

$$\begin{aligned} E_t [r_{i,t+1} - r_f] &= -\frac{1}{2} \text{var}_t (r_{i,t+1}^{net}) + E_t [c_{i,t+1}] + \gamma_t [\text{cov}_t (r_{i,t+1}, r_{W,t+1}) \\ &\quad + \text{cov}_t (c_{i,t+1}, c_{W,t+1}) - \text{cov}_t (c_{i,t+1}, r_{W,t+1}) - \text{cov}_t (r_{i,t+1}, c_{W,t+1})]. \end{aligned} \quad (2)$$

The asset pricing equation of investable securities is similar to the liquidity-adjusted CAPM of AP but the market risk premium and liquidity risk premiums are aggregated at the world level and with the addition of one-half the variance of excess net returns. Hence, as in BHL local liquidity variables affect expected returns even under full integration. The asset pricing equation under full market integration of BHL is not identical to ours as BHL assumes that liquidity is a second risk factor that affects the world pricing kernel in addition to the world market risk factor and hence triggers its own price of liquidity risk. Except for the Jensen's inequality term, our pricing equation for investables is the extension of AP to an international setup.

2.3.2 Non-investable set

For a non-investable security $n \in \{1, \dots, N_N\}$, the expected net excess return at equilibrium is given by,

$$\begin{aligned} E_t [r_{n,t+1}^{net} - r_f] + \frac{1}{2} \text{var}_t (r_{n,t+1}^{net}) &= \gamma_t \text{cov}_t (r_{n,t+1}^{net}, r_{W,t+1}^{net}) \\ &\quad + \pi_t \text{cov}_t (r_{n,t+1}^{net}, r_{N,t+1}^{net} \mid \mathbf{r}_{i,t+1}^{net}) \end{aligned} \quad (3)$$

where $\pi_t = \left(\frac{\gamma^f}{W_t^f} - \frac{\gamma_t}{W_t} \right) M_{N,t}$ is the price of local conditional risk. We then express the covariances in terms of gross returns

$$\begin{aligned} cov_t(r_{n,t+1}^{net}, r_{W,t+1}^{net}) &= cov_t(r_{n,t+1}, r_{W,t+1}) + cov_t(c_{n,t+1}, c_{W,t+1}) \\ &\quad - cov_t(c_{n,t+1}, r_{W,t+1}) - cov_t(r_{n,t+1}, c_{W,t+1}), \end{aligned} \quad (4)$$

$$\begin{aligned} cov_t(r_{n,t+1}^{net}, r_{N,t+1}^{net} | \mathbf{r}_{i,t+1}^{net}) &= cov_t(r_{n,t+1}, r_{N,t+1} | \mathbf{r}_{i,t+1}^{net}) + cov_t(c_{n,t+1}, c_{N,t+1} | \mathbf{r}_{i,t+1}^{net}) \\ &\quad - cov_t(c_{n,t+1}, r_{N,t+1} | \mathbf{r}_{i,t+1}^{net}) - cov_t(r_{n,t+1}, c_{N,t+1} | \mathbf{r}_{i,t+1}^{net}). \end{aligned} \quad (5)$$

Substituting (4) and (5) in (3) we obtain,

$$\begin{aligned} E_t[r_{n,t+1} - r_f] &= -\frac{1}{2} var_t(r_{n,t+1}^{net}) + E_t[c_{n,t+1}] + \gamma_t [cov_t(r_{n,t+1}, r_{W,t+1}) + cov_t(c_{n,t+1}, c_{W,t+1}) \\ &\quad - cov_t(c_{n,t+1}, r_{W,t+1}) - cov_t(r_{n,t+1}, c_{W,t+1})] \\ &\quad + \pi_t [cov_t(r_{n,t+1}, r_{N,t+1} | \mathbf{r}_{i,t+1}^{net}) + cov_t(c_{n,t+1}, c_{N,t+1} | \mathbf{r}_{i,t+1}^{net}) \\ &\quad - cov_t(c_{n,t+1}, r_{N,t+1} | \mathbf{r}_{i,t+1}^{net}) - cov_t(r_{n,t+1}, c_{N,t+1} | \mathbf{r}_{i,t+1}^{net})]. \end{aligned} \quad (6)$$

The extra conditional liquidity risks are

1. $cov_t(c_{n,t+1}, c_{N,t+1} | \mathbf{r}_{i,t+1}^{net})$: the conditional commonality in transaction risk between a security n and the local non-investable market N ;
2. $cov_t(c_{n,t+1}, r_{N,t+1} | \mathbf{r}_{i,t+1}^{net})$: the conditional covariance between a security's transaction cost and the local non-investable market return;
3. $cov_t(r_{n,t+1}, c_{N,t+1} | \mathbf{r}_{i,t+1}^{net})$: the conditional covariance between a security's return and the local non-investable market transaction cost.

As in AP, (1) affects required returns positively because investors want to be compensated for holding securities with higher transaction costs when the local market transaction costs increase; (2) and (3) affect required returns negatively because investors are willing to accept a lower return on an asset with lower transaction costs in a down local market, or an asset with higher return when local market transaction costs are high.

Note that the equilibrium asset pricing model implies that global risk is measured vis-à-vis the world market return (and aggregate liquidity) as the total world market portfolio is the correct portfolio to measure risk sharing.

To gain further insight into the price of conditional risk, we write the conditional covariance as,

$$\text{cov}_t(r_{n,t+1}^{net}, r_{N,t+1}^{net} | \mathbf{r}_{i,t+1}^{net}) = \text{cov}_t(r_{n,t+1}^{net}, r_{N,t+1}^{net}) - \text{cov}_t(r_{n,t+1}^{net}, r_{DP,t+1}^{net}) \quad (7)$$

where $r_{DP,t+1}^{net}$ is the net return on the diversification portfolio DP with weights $\omega_{DP,t} = \Sigma_{ii,t+1}^{-1} \Sigma_{in,t+1} \mathbf{M}_{n,t} / M_{N,t}$, which is the portfolio of investable assets that is most highly correlated with the market portfolio of non-investable securities. The DP portfolio is the one held long by the domestic investor as the best substitute for the non-investable segment, but short by the foreign investor to reduce his local risk exposure.

If the correlation between returns on the value-weighted market portfolio of non-investable securities, $r_{N,t}$, and its diversification portfolio, $\rho_{N,DP,t}$, is one, the market is effectively integrated, all of the conditional market risk premium and conditional liquidity risk premiums disappear and only global market and liquidity risk are priced as in AP. Hence perfect spanning would eliminate the emerging market valuation discount due to illiquidity risk but would not eliminate the global premiums or discounts that result from co-variation of the asset return and its transaction cost with the global market return and global transaction costs. Under complete segmentation, only local market and local liquidity risks are priced. Under partial segmentation, both global and conditional local market return and liquidity risks are priced.

We can capture the degree of segmentation of the foreign market by defining the integration index $II_t = \rho_{N,DP,t}^2$. By aggregating Equation (7) over the set of non-investable securities we obtain,

$$\text{var}_t(r_{N,t+1}^{net} | \mathbf{r}_{i,t+1}^{net}) = \text{var}_t(r_{N,t+1}^{net}) (1 - II_t). \quad (8)$$

We can then obtain the conditional expected gross excess return on the local non-investable market portfolio, $r_{N,t+1}$ as,

$$\begin{aligned}
E_t[r_{N,t+1} - r_f] &= -\frac{1}{2}var_t(r_{N,t+1}^{net}) + E_t[c_{N,t+1}] + \gamma_t[cov_t(r_{N,t+1}, r_{W,t+1}) + cov_t(c_{N,t+1}, c_{W,t+1}) \\
&\quad - cov_t(c_{N,t+1}, r_{W,t+1}) - cov_t(r_{N,t+1}, c_{W,t+1})] \\
&\quad + \pi_t[var_t(r_{N,t+1}) + var_t(c_{N,t+1}) - 2cov_t(r_{N,t+1}, c_{N,t+1})](1 - II_t). \quad (9)
\end{aligned}$$

3 Econometric specification

Our empirical methodology follows three main steps. In a first step, we estimate univariate GARCH models and copula models to obtain, respectively, variances and covariances needed to estimate the time-varying weights of the diversification portfolio. In a second step, we estimate the time-varying price of world market risk and the coefficients of the time-varying world variance. In the third step, we impose these estimates in the country estimations. For each country, we have a system of six equations for the gross returns and the transaction costs of the world market portfolio, the non-investable stock market portfolio, and its diversification portfolio. We then use the multivariate generalized autoregressive conditional heteroskedasticity (GARCH)-in-mean methodology with a diagonal covariance specification (see [Bollerslev, Engle, and Wooldridge \(1988\)](#)). The estimation procedure results in sampling errors from the two first stages. However, imposing the world price of risk in the third step is in line with the theoretical model restrictions and produces more powerful tests.

3.1 Estimation of the diversification portfolio

The diversification portfolio for country k is composed of the value-weighted world market portfolio W , the value-weighted portfolio of investable securities I^k , the value-weighted portfolio of substitute assets S^k which includes ADRs, GDRs, and direct listings, and all its Country Funds and related Exchange Traded Funds, which we denote as $CF^k = \{CF_1^k, CF_2^k, \dots\}$. We discuss in Section 4 our construction methodology for each of these portfolios. We form a dynamic replicating portfolio

whose allocation at time t uses conditional covariances:

$$\omega_{DP^k,t} = \Sigma_{(W,I^k,S^k,CF^k),t}^{-1} \Sigma_{N^k-(W,I^k,S^k,CF^k),t} \quad (10)$$

where $\Sigma_{(W,I^k,S^k,CF^k),t}$ is the covariance matrix at time t of the vector

$$\mathbf{r}_{(W,I^k,S^k,CF^k),t}^{net} = \left[r_{W,t}^{net}, r_{I^k,t}^{net}, r_{S^k,t}^{net}, r_{CF_1^k,t}^{net}, r_{CF_2^k,t}^{net}, \dots \right]^\top$$

and $\Sigma_{N^k-(W,I^k,S^k,CF^k),t}$ is the vector of covariances between the net return on the value-weighted portfolio of non-investable securities $r_{N^k,t}^{net}$ and the vector $\mathbf{r}_{(W,I^k,S^k,CF^k),t}^{net}$.

We estimate one volatility model for each asset and then obtain the correlations from a dynamic normal copula (DNC) estimated on the residuals as in [Christoffersen and Langlois \(2013\)](#). All model parameters are estimated by maximum likelihood. For each asset, we estimate an AR(P)-NGARCH model on net returns,

$$\begin{aligned} r_{j,t+1} - c_{j,t+1} &= \phi_{j,0} + \sum_{p=1}^4 \phi_{j,p} (r_{j,t+1-p} - c_{j,t+1-p}) + \epsilon_{j,t+1}, \quad \epsilon_{j,t+1} \sim N(0, \sigma_{j,t+1}^2) \\ \sigma_{j,t+1}^2 &= \bar{\sigma}_j (1 - b_j - a_j (1 + c_j^2)) + b_j \sigma_{j,t}^2 + a_j (\epsilon_{j,t} - \sigma_{j,t} c_j)^2 \end{aligned}$$

where $j \in \{W, N^k, I^k, S^k, CF_1^k, CF_2^k, \dots\}$ and $b_j + a_j (1 + c_j^2) < 1$. We have found that an AR(4) for the expected value of $r_{j,t+1} - c_{j,t+1}$ was sufficient to capture the auto-correlation in the transaction costs.

Then, we estimate one DNC model in which the dynamic correlations are driven by the dynamic,

$$\Gamma_{t+1} = \Omega_C (1 - b_C - a_C) + b_C \Gamma_t + a_C u_t u_t^\top$$

where Ω_C is the long-run correlation matrix, b_C and a_C are the correlation dynamic parameters with $b_C + a_C < 1$, and $u_{j,t} = \frac{\epsilon_{j,t}}{\sqrt{\Gamma_{j,j,t} \bar{\sigma}_{j,t}^2}}$ are the standardized residuals with the [Aielli \(2013\)](#) correction. We obtain the dynamic correlation through the standardization $C_{j,l,t} = \frac{\Gamma_{j,l,t}}{\sqrt{\Gamma_{j,j,t} \Gamma_{l,l,t}}}$.

Estimating conditional covariances is complicated by the fact that the replicating assets have different time series length, with some series not overlapping at all. To ensure positive definiteness of the covariance matrix of $r_{N^k,t}$ and $r_{(W,I^k,S^k,CF^k),t}$, we estimate the DNC model by maximizing

the log-likelihood using all available assets at each time period. This allows us to estimate one correlation model on a unbalanced panel of asset returns. Once an asset j becomes available, we set its first conditional correlations with other available assets l to its long-run parameter $\Omega_{j,l}$. We examine the unbiasedness of the estimators via simulations. The results suggest that the estimator of the DNC with missing values in the panel is unbiased.

Since the distribution of the GARCH shocks $\epsilon_{j,t}$ and the copula is gaussian, the joint distribution obtained by combining them is also multivariate gaussian. Hence, we use the conditional variances obtained from the AR(4)-NGARCH models and the conditional correlations obtained from the DNC model to obtain the conditional covariances needed to estimate the diversification portfolio weights.

3.2 Asset pricing tests

We test the conditional asset pricing model for each country k . Specifically, we estimate the following system of equations,

$$\begin{aligned}
r_{W,t+1} - r_{f,t+1} &= \alpha_W - \frac{1}{2}var_t(r_{W,t+1}^{net}) + \kappa_W E_t[c_{W,t+1}] + \gamma_{t+1}var_t(r_{W,t+1}^{net}) + \epsilon_{1,t+1} \\
r_{N^k,t+1} - r_{f,t+1} &= \alpha_{N^k} - \frac{1}{2}var_t(r_{N^k,t+1}^{net}) + \kappa_{N^k} E_t[c_{N^k,t+1}] + \gamma_{t+1}cov_t(r_{N^k,t+1}^{net}, r_{W,t+1}^{net}) \\
&\quad + \pi_{t+1}^k var_t(r_{N^k,t+1}^{net}) \left(1 - \frac{cov_t(r_{N^k,t+1}^{net}, r_{DP^k,t+1}^{net})^2}{var_t(r_{N^k,t+1}^{net}) var_t(r_{DP^k,t+1}^{net})} \right) + \epsilon_{2,t+1} \\
r_{DP^k,t+1} - r_{f,t+1} &= \alpha_{DP^k} - \frac{1}{2}var_t(r_{DP^k,t+1}^{net}) + \kappa_{DP^k} E_t[c_{DP^k,t+1}] \\
&\quad + \gamma_{t+1}cov_t(r_{DP^k,t+1}^{net}, r_{W,t+1}^{net}) + \epsilon_{3,t+1} \\
c_{W,t+1} &= E_t[c_{W,t+1}] + \epsilon_{4,t+1} \\
c_{N^k,t+1} &= E_t[c_{N^k,t+1}] + \epsilon_{5,t+1} \\
c_{DP^k,t+1} &= E_t[c_{DP^k,t+1}] + \epsilon_{6,t+1}
\end{aligned} \tag{11}$$

where the variances and covariances are obtained from the covariance matrix H_t of the 6×1 vector $\epsilon = (\epsilon_{1,t}, \dots, \epsilon_{6,t})^\top$. We allow for intercepts $\alpha_j, j \in \{W, N^K, DP^K\}$ in the estimation but the model implies that the intercepts are zero. The expected transaction costs are modeled using an AR(4) as $E_t[c_{j,t+1}] = \phi_{j,0} + \sum_{p=1}^4 \phi_{j,p} c_{j,t+1-p}$.

In the next section, we use the bid-ask spread measure of [Corwin and Schultz \(2012\)](#) estimated from the high and low prices as transaction cost $c_{j,t}$. To test our model, we estimate three specifications. First, we impose the model-implied constraint that $\kappa_W = \kappa_{Nk} = \kappa_{DPk} = 1$ and use it as our main specification. Admittedly excess returns net of bid-ask spreads overestimate the true cost of trading. The asset pricing model assumes that all investors have the same horizon, an assumption unlikely to be verified in reality. In a second specification discussed in [Section 6](#), we follow AP and set the coefficients of the expected transaction cost on the right hand side to the average turnover values to capture the fact that the effect of expected transaction costs in the data may differ from our stylized model¹⁴. In a third specification, we estimate the model in gross returns where the liquidity costs do not enter the first three equations in (11). This is the asset pricing model that we use as the benchmark specification.

We allow the price of world market risk to change through time as suggested in the literature (see, among others [Harvey, 1991](#); [de Santis and Gerard, 1997](#)). In order to estimate the model, we need to specify the functional form of the time-varying price of world covariance risk. Because the model implies that the price of world market risk should be positive, we model its dynamics using a square function of the global instruments $\mathbf{Z}_{W,t} = (1, \mathbf{Z}_{W,t}^{*\top})^\top$,

$$\gamma_{t+1} = (\lambda_W^\top \mathbf{Z}_{W,t})^2. \quad (12)$$

If the world market risk is priced, we should reject the hypothesis that the $\lambda_{j,W} = 0$, for $j \geq 1$. If the world market risk is time-varying, we should reject that $\lambda_{j,W} = 0$, for $j > 1$.

Similarly, we allow the price of conditional market risk to be time-varying. Because the price of conditional market risk should be positive, we model the price dynamics using a square function of the local instruments $\mathbf{Z}_{k,t} = (1, \mathbf{Z}_{k,t}^{*\top})^\top$,

$$\pi_{t+1}^k = (\lambda_k^\top \mathbf{Z}_{k,t})^2.$$

If local risk is priced, we should reject the hypothesis that the $\lambda_{j,k} = 0$, for $j \geq 1$. If local risk is

¹⁴Volume data are not available for as long a time period as our bid-ask spread measure. We compute the value-weighted average of turnover for all stocks in our portfolios whenever volume data becomes available. Average turnover values are reported in [Table 3](#) when we discuss illiquidity risk premiums.

time-varying, we should reject the hypothesis that the $\lambda_{j,k} = 0$, for $j > 1$.

The choice of instruments used in the parametrization of the prices of risk follow the literature (see, among others [Ferson and Harvey, 1993, 1994](#); [Dumas and Solnik, 1995](#); [de Santis and Gerard, 1997](#)). The global instruments $\mathbf{Z}_{W,t}^{*\top}$ include the world dividend yield in excess of the risk free rate, the change in the U.S. term premium, measured by the yield difference between the three-month T-bill and the 10-year T-bond, and the U.S. default premium measured by the yield difference between Moody's Baa and Aaa rated bonds. The local instruments $\mathbf{Z}_{k,t}^{*\top}$ include the local equity return in excess of the risk free rate, the local excess dividend yield and the change in local nominal exchange rate.

We use the multivariate generalized autoregressive conditional heteroskedasticity (GARCH)-in-mean methodology with a diagonal covariance specification (see [Bollerslev et al., 1988](#)), which assumes that the variances depend only on past squared residuals and an autoregressive component, while the covariances depend on the past cross-product of residuals and an autoregressive component. The dynamics of the conditional covariances H_t are specified as,

$$H_t = H_0 \circ (\boldsymbol{\iota}^\top - \mathbf{b}\mathbf{b}^\top - \mathbf{a}\mathbf{a}^\top) + \mathbf{b}\mathbf{b}^\top \circ H_{t-1} + \mathbf{a}\mathbf{a}^\top \circ \boldsymbol{\epsilon}_{t-1}\boldsymbol{\epsilon}_{t-1}^\top \quad (13)$$

where $\boldsymbol{\iota}$ is a (6×1) vector of ones, $\mathbf{a}$ and $\mathbf{b}$ are (6×1) vectors of unknown parameters, $\circ$ is the Hadamard (element by element) matrix product, and each element of $\mathbf{b}\mathbf{b}^\top + \mathbf{a}\mathbf{a}^\top$ is smaller than one. The advantage of this multivariate GARCH-in-mean parameterization is that it ensures positive definiteness of the covariance matrix while reducing the number of parameters to be estimated.

We estimate the model and compute the parameters' covariance matrix by the quasi-maximum likelihood (QML) of [Bollerslev and Wooldridge \(1992\)](#), using the BFGS ([Shanno \(1985\)](#)) algorithm to update the Hessian.

Since the theory predicts that the world price of risk should be the same for each country, we first estimate the world equation in the system of Equations (11) to obtain the time-varying world price of risk and the coefficients of the time-varying world variance (see, for example, [Bekaert and Harvey, 1995](#)). We then impose the estimates $\hat{\alpha}_W$, $\hat{\kappa}_W$, $\hat{E}_t[c_{W,t+1}]$, $\hat{\gamma}_{t+1}$, $\hat{H}_{W,W,0}$, $\hat{a}_W$, and $\hat{b}_W$ in the country estimations.

4 Data

We include all emerging markets that were at some point in the MSCI Emerging Market index (Argentina, Brazil, Chile, China, Colombia, Czech Republic, Egypt, Greece, Hungary, India, Indonesia, Israel, Jordan, Malaysia, Mexico, Morocco, Pakistan, Peru, Philippines, Poland, Portugal, Qatar, Russian Federation, South Africa, South Korea, Sri Lanka, Taiwan, Thailand, Turkey, United Arab Emirates, and Venezuela). For each of these markets, we obtain from Datastream daily low, high, and closing prices, trading volumes, total returns, and monthly market capitalizations in both local and U.S. currencies for all securities classified as equity, American Depositary Receipts, and Global Depositary Receipts. To test our asset pricing model, we use daily data to build our monthly liquidity measures and U.S. dollar denominated returns in excess of the one-month Treasury bill rate obtained from Kenneth French’s website. We use the longest times series available for each security, and our sample ends in December 2015.

We apply the filters on daily and monthly data listed in Appendix A. As pointed out in [Ince and Porter \(2006\)](#), Datastream data has to be handled with care, and we follow [Lee \(2011\)](#) and [Karolyi et al. \(2012\)](#) in applying filters to the data.

To test our asset pricing model, we need direct measures of transaction costs. For all stocks, we compute the bid-ask spread measure of [Corwin and Schultz \(2012\)](#) to approximate transaction costs. Using two consecutive days, [Corwin and Schultz \(2012\)](#) calculate both daily high-low price ranges and the high-low price range over the two days to distinguish the bid-ask spread from daily variance. By applying their methodology, we obtain daily estimates of the bid-ask spread. Following their suggestion, we set negative daily spreads to zero. The monthly spread measure is the average of daily measures within the month for months in which there are at least ten daily spreads available. While it is a good proxy for the cost of a round-trip transaction of small size, this measure does not capture other dimensions of liquidity such as market depth, resilience, and search costs.

The non-investable and investable portfolios for each country contain only local stocks. We define a local stock as one traded on the major stock exchange in each country, which is the one with the highest number of equities listed. Exceptions to this rule are Brazil (Rio de Janeiro and Bovespa), China (Hong Kong, Shanghai, and Shenzhen), Germany (Frankfurt and XETRA),

Japan (Osaka and Tokyo), Russia (Moscow and MICEX), South Korea (KOSDAQ and Korea Stock Exchange), and United States (NASDAQ, NYSE, and NYSE Arca). We test our asset pricing model on emerging markets, but use stock exchanges from developed markets to determine Depositary Receipts' (DR) and direct listings' eligibility and to build the world market portfolio as discussed below. To ensure we have only local common stocks, we apply the filters from [Griffin, Kelly, and Nardari \(2010\)](#) on local stocks' names and industries.¹⁵

Each month, we build value-weighted return and bid-ask spread indices. We construct one set of indices that contains all stocks for a given country, one that contains only investable stocks, and one that contains only non-investable stocks. The compositions of our portfolios change every month. To be eligible, a stock must have over the last three months valid U.S. dollar denominated monthly returns, market capitalizations, and bid-ask spreads. Then, we follow [Karolyi and Wu \(2012\)](#) and characterize a stock as investable if there exists a related cross-listed stock, an ADR, or a GDR traded on a major stock exchange in a developed country with a valid monthly return. We identify cross-listing stocks using ISINs. We identify ADRs and GDRs by retrieving from Bloomberg their underlying SEDOLs and ISINs.¹⁶ We also use country specific filters; for example B shares in China and the Philippines are classified as investable.

Alternative determinants of investability include the investability weight factor (IWF) computed by S&P/IFC or the foreign inclusion factor (FIF) computed by MSCI. In unreported results, we obtain from MSCI the FIFs for mid- and large-cap companies from 1995 to 2015 and additionally require that more than half of the market capitalization company be investable ($FIF_i \geq 0.5$) for a stock to be investable. We found no difference in classification. Another investability proxy is the foreign ownership holdings used in [Errunza and Ta \(2015\)](#). The foreign ownership holdings data from Lionshare start in 1999 and do not change the sample of investable stocks compared to the existence of depository receipts or direct listings as determinant of the investability of a stock as in [Karolyi and Wu \(2012\)](#). [Bekaert, Harvey, Lundblad, and Siegel \(2011\)](#) and [Carrieri, Chaieb, and Errunza \(2013\)](#) show that implicit barriers to foreign investments results in investable stocks not being effectively integrated with the world economy. Therefore, we use in Section 6 the portfolio of all local stocks as our non-investable set as a robustness check.

¹⁵We add "BDR" to the list of keywords to remove Brazilian DRs.

¹⁶We are able to match 2,950 of the 3,729 DRs using Bloomberg

We use all eligible local stocks listed in 53 developed and emerging markets to build the value-weighted world market portfolio. Then, for each country we build a portfolio of substitute assets, S^k , that contains all eligible ADRs, GDRs, and direct listings. The objective is to obtain the best replicating asset for the portfolio of local non-investable stocks. Hence, for each stock for which there exists a DR or a direct listing, we form a value-weighted portfolio of all DRs and direct listings using their respective market capitalizations. The weight of each of these portfolios in the portfolio of substitute assets S^k is proportional to the market capitalization of their respective local stock. Hence, we obtain a portfolio that is as close as possible to the non-investable portfolio of local stocks and that allocates more to larger DRs and direct listings. We also use a list of Exchange Traded Funds (ETF) and country funds (CF) for each country mainly compiled from Morningstar, Bloomberg, iShares, and CRSP. These assets will be used to replicate the portfolio of non-investable assets. Note that when we form the diversification portfolio, we use as replicating assets the world market portfolio W , the portfolio of investable stocks I^k , the portfolio of ADRs, GDRs, and direct listings S^k , and all available country funds and ETFs $CF_1^k, CF_2^k, \dots$. When we use in Section 6 as a robustness check the portfolio of all local stocks as the set of non-investable securities, we remove the portfolio of investable stocks from the set of replicating assets.

To be able to estimate the dynamics of variances and covariances, we only keep countries for which we have at least a 10-year time series for both the non-investable and the investable index. Our final sample is comprised of the following 21 countries: Argentina, Brazil, Chile, China, Egypt, Greece, Hungary, India, Indonesia, Israel, Malaysia, Mexico, Morocco, Philippines, Poland, Russia, South Africa, South Korea, Taiwan, Thailand, and Turkey. Further, we require at least a 10-year time series for replicating assets to be included in the diversification portfolio.

4.1 Summary statistics

Table 1 reports the number of stocks in the sample every five years from 1988 to 2015. The table also reports the descriptive statistics of the monthly returns and the [Corwin and Schultz \(2012\)](#) bid-ask spread measure.

The number of stocks varies across countries and years as a result of data coverage and the set of filters detailed in Appendix A. It is quite small at the beginning of the period but increases steadily. Table 1 also shows the fraction of investables reported in column (1) in italics. There

are 615 stocks in 1988 of which 1% are investable and 8,513 stocks at the end of the sample in December 2015 with 6% investable. The proportion of investable stocks steadily increases until the mid-2000s, but has stagnated since. In 2015, China is the country with the largest number of stocks in the sample with 2,218 stocks of which 183 are investables, while Morocco has the smallest number of stocks with only 14 stocks that are essentially non-investable. Hungary and Mexico are the countries with the largest fraction of investables standing at 47% (9 over 19) and 51% (39 over 77), respectively. In the country estimations, we use the longest sample period in which the non-investable portfolio is available, and therefore the starting date varies across countries.

The last four columns of Table 1 show averages and standard deviations of the monthly excess returns and of the bid-ask spread measure. The return figures are annualized and for comparison purposes the statistics are computed over the sample period during which both the non-investable and the investable index are available. The U.S. dollar monthly returns are in excess of the 1-month U.S. T-bill rate. The emerging markets display high average returns but also high volatility as well-documented in the literature. Annualized equity volatilities range between 17% and 65%.

[Insert Table 1]

We observe striking cross-country differences in the value-weighted average illiquidity of the emerging market indexes. According to the bid-ask spread measure, India is on average the least liquid, while Chile and Morocco are the most liquid. We report in italics in column (11) of Table 1, the ratio of the average bid-ask spread for non-investables to the average bid-ask spread for investables. The ratio is above one for all but three countries and it reaches 1.63 for Philippines, which indicates that non-investables are more illiquid than investables. To further examine the difference in liquidity measures and their dynamics across investables and non-investables, we report in Figure 1 value- and equal-weighted averages of bid-ask spread measures across countries for portfolios of non-investable stocks (left column) and investable stocks (right column). In each graph, the horizontal line reports the time-series average. The plots show interesting differences across investables and non-investables. Investables are most of the time and on average more liquid than non-investables. There is also no clear downward trend over time in bid-ask spreads for both investables and non-investables. The plots also show that illiquidity varied significantly over time.

We report in this and all following figures gray areas indicating NBER recessions (July 1990 to

March 1991, March to November 2001, and December 2007 to June 2009), the European exchange rate mechanism (ERM) crisis (September 1992 to August 1993), the Tequila crisis (December 1994 to January 1995), the East Asia crisis (June to December 1997), the Russian default and Long-Term Capital Management crisis (August to December 1998), and the Euro-sovereign debt crisis (January 2010 to December 2012). Other important events, such as the bankruptcy of Lehman Brothers (September 2008) and the U.S. debt downgrade (August 2011) coincide with these other events. The impact of the Asian, the Russian default, the Long Term Capital Management (LTCM), and the subprime crises on bid-ask spreads is clear. The rise in illiquidity during the 2008-2009 crisis is quite strong for both investables and non-investables. After the global financial crisis, illiquidity reverted back to average levels for both investables and non-investables.

[Insert Figure 1]

5 Results

In this section, we estimate our asset pricing model and test whether world market and conditional market risk factors are priced and are time-varying. We also examine the economic contribution and the dynamics of the different risk premia, specifically the global and local liquidity premia. We do this by running the multivariate GARCH-in-mean on the world market portfolio, the EM non-investables and their diversification portfolios as detailed in Section 3.

5.1 Prices of risk

First we estimate the price of world market risk in a conditional set up allowing the price to vary over time ($Z_{W,t} = (\mathbf{1}, Z_{W,t}^{*'})'$). We test the world market model using the three model specifications; (1) the main specification in which we use covariances of net returns and we impose the model-implied restriction $\kappa = 1$, (2) the alternative specification in which we use covariances of net returns and calibrate κ to the average monthly turnover, and (3) the benchmark specification in which we use covariance of gross returns and set κ to zero in the system of Equations 11.

Panel A of Table 2 shows that the price of world market risk is statistically significant and time-varying at the 1% level. The last three columns report the pricing errors α_W . The estimation of the liquidity-adjusted model results in a reduction of the pricing error from 0.66% for the benchmark

specification to 0.06% for the main specification. Figure 2 reports the time varying price of world market risk. The price of world market risk increases during economic contractions and crises periods.¹⁷ The average estimates of aggregate risk aversion from the three specifications are 1.15, 1.29, and 1.45 for the main, alternative and benchmark specifications, respectively. The price of world market risk is equal to the aggregate risk aversion. Hence, these values are economically meaningful.

[Insert Figure 2]

For the local prices of risk we report and discuss the main and benchmark specifications. Figure 3 plots the median, 25%, and 75% of the local prices of risk for our EM sample obtained from the estimation of our main specification. The prices fluctuate through time and across countries. They tend to increase during economic recessions and financial crises. The figure also reports the average local price across the 21 countries and across the sample period. The cross-country average price of local risk is 3.6. The price of local risk is proportional to the differential in risk aversion between an EM average investor and the world average investor and hence the size seems economically plausible.

[Insert Figure 3]

Panel B of Table 2 contains a number of specification tests. For each country, we report in columns (2)-(5) robust Wald test statistics and their p -value for the significance and time variation in the prices of conditional local market risk¹⁸. It is important to note that the p -values we report ignore the sampling error associated with the first two step estimations. Our main specification shows that the price of local market risk is significant and time varying for all EM countries. The test statistics for the benchmark specification with gross returns, reported in columns (3) and (5), confirm these results.

Columns (6)-(9) in Panel B of Table 2 report pricing errors for the market portfolio of noninvestables, α_{Nk} , and its diversification portfolio, α_{DPk} . Each row contains the estimates and their

¹⁷The period of contraction is measured from peak to trough as determined by the National Bureau of Economic Research, NBER. The crises periods include the Asian, Tequila, LTCM, and global financial crisis.

¹⁸The unreported estimates on coefficients used to parameterize the prices of risk are statistically significant in most cases. The results are available from the authors.

t -ratios underneath. We report pricing errors for our main specification in columns (6) and (7) and pricing errors for the benchmark specification in columns (8) and (9).

The main specification has one significant pricing error (in bold and red) whereas there are 12 significant pricing error for the benchmark specification. Notice that both the main and benchmark specifications have the same number of factors and hence same degrees of freedom. The only difference is the liquidity adjustment. Overall, the results lend support to the net return model.

We do not report the parameter estimates ($\mathbf{a}$ and $\mathbf{b}$) for the conditional covariance processes, but these are significant in all cases, of the usual size, and satisfy the stationarity condition, i.e., each element of $\mathbf{b}\mathbf{b}^\top + \mathbf{a}\mathbf{a}^\top$ is smaller than one. This value is also close to one in most instances, suggesting that the variance and covariance processes in Σ_t are highly persistent.

[Insert Table 2]

5.2 Liquidity risk premium

We examine the economic contribution of the level, world risk, and conditional local risk premia coming from illiquidity costs. Following AP, we use the alternative specification with the κ s calibrated to the average monthly turnover of each portfolio. For each EM, we compute the time-series of the illiquidity expected return component as,

$$\kappa_{N^k} E_t [c_{N^k,t+1}]. \quad (14)$$

Then, we compute the expected return component coming from the world risk premium as,

$$\gamma_{t+1} (cov_t(c_{N^k,t+1}, c_{W,t+1}) - cov_t(r_{N^k,t+1}, c_{W,t+1}) - cov_t(r_{N^k,t+1}, r_{W,t+1})), \quad (15)$$

and finally the expected return component coming from the conditional local risk premium as,

$$\begin{aligned} \pi_{t+1}^k (var_t(c_{N^k,t+1}) - 2cov_t(r_{N^k,t+1}, c_{N^k,t+1}) + cov_t(r_{N^k,t+1}, c_{DP,t+1}) + cov_t(c_{N^k,t+1}, r_{DP,t+1}) \\ - cov_t(c_{N^k,t+1}, c_{DP,t+1})). \end{aligned} \quad (16)$$

Table 3 reports the average turnover and the time-series average, minimum, and maximum of

each component as well as the total expected return coming from illiquidity terms. Column (1) in Table 3 reports the annualized average monthly turnover which range from 1.21% to 25.75% across EMs. While both world illiquidity and local illiquidity risk premium components in columns (5) and (8) are small on average, they vary significantly as illustrated by the wide range of minima and maxima in columns (6), (7), (9), and (10). The risks involving only transaction costs, $cov_t(c_{N,t+1}, c_{W,t+1})$, $var_t(c_{N,t+1})$, and $cov_t(c_{N,t+1}, c_{DP,t+1})$, (not reported) are orders of magnitude smaller than other variances and covariances and marginally contribute to the total risk premium. This is in line with AP who report that the commonality in liquidity is the smallest contributor. Overall, the illiquidity level is the biggest contributor on average to the expected return illiquidity component. Column (11) reports the time-series average of the sum of all illiquidity components, which averages 0.75% per year on average across countries. This average is lower than the 4.6% reported in AP. Whereas they form 25 portfolios sorted on liquidity level and examine the spread between the most illiquid and the most liquid portfolio, we compute the average illiquidity risk premium at the value-weighted market level which puts less weight on small and illiquid stocks.

[Insert Table 3]

Figures 4 report for each EM the total illiquidity premiums over time. The time-series variation are striking; the impact of the financial crisis is clear in most graphs. Consistent with past findings, world liquidity risk is small on average but increases during crisis periods.

[Insert Figures 4]

6 Robustness

Past studies (see, for example, [Carrieri et al., 2013](#)) suggest that although investables are freely available to non-local investor they still command a local risk premium. We therefore assume that the whole stock market of the emerging country is non-investable. We form the diversification portfolios for the local market portfolio that includes both investables and non-investables and estimate the system of Equations (11) with the world market portfolio, the portfolio of local stocks, and its diversification portfolio. Table 4 summarizes the hypotheses tests of significant and time-varying conditional local price of risk from the main and benchmark specifications and reports the

intercepts of the market portfolio of non-investables and its DP from both model specifications. The local price of risk is significant and time-varying in all cases and across the two specifications. The intercepts from the main specification are insignificant compared to the benchmark specification where 16 of the pricing errors are significant. The relative contribution of the liquidity risk premiums are unreported and available from the authors. The magnitudes are similar to the estimation with investables and non-investables and confirm that illiquidity risk premium is valued more in times of financial market stress. All of these findings lend strong support to our liquidity asset pricing model in partially segmented markets.

7 Conclusion

We develop a formal international asset pricing model that takes into account cross-border investment barriers as well as random transaction costs to analyze the effects of illiquidity cost and systematic liquidity risk factors on the pricing of EM securities. In our model, the freely traded securities command a global market risk premium and world liquidity premia whereas the securities that can be held by only a subset of investors command additionally a conditional local market risk premium and conditional local liquidity risk premia.

We estimate the model for 21 EMs. Empirical test results support theoretical predictions. Specifically, the price of world market risk is positive, statistically, and economically significant. The cross-country average price of local risk is 3.6, however, the magnitude of the price of local risk varies across countries. Although, on average, both the world and local liquidity risk premia are quite small their importance increases during crisis periods corroborating past evidence. Our results are not sensitive to alternative model specification with unrestricted expected liquidity costs, the choice of the set of non-investables, and the investability criteria.

In conclusion, our model provides a formal framework for testing important issues such as liquidity levels and risks in a realistic world market setting and for examining the interaction between investability and illiquidity.

Appendix A - Data filters and corrections

We apply different filters on Datastream daily and monthly data:

- All datatypes do not have the same time series length on Datastream. For all datatypes, we keep only the sample period for which the unadjusted price of the security is available.
- Following [Lee \(2011\)](#) and [Lesmond \(2005\)](#), we drop days for which more than 90% of local stocks in a given country have zero returns on that day.
- When the return over two consecutive days is smaller than 50%, but one of the daily returns is larger than 100%, we remove both daily returns.
- Similarly, when the return over two consecutive months is smaller than 50%, but one of the monthly returns is larger than 300%, we remove both monthly returns.
- To ensure that our monthly returns are not impacted by rounding, we specify the number of decimals to 9 when converting the total return indexes to U.S. dollar using the function $DPL\#(X(RI) \text{ U}\$, 9)$.
- Following [Corwin and Schultz \(2012\)](#), we remove daily data for which:
 1. the ratio of high to low price is smaller than 1 or larger than 8,
 2. any of the low, high, and closing price both for the day and the previous day are unavailable.

We remove monthly bid-ask spread measure larger than 100%.

References

- Acharya, V., and L. H. Pedersen. 2005. Asset pricing with liquidity risk. *Journal of Financial Economics* 77:375–410.
- Aielli, G. P. 2013. Dynamic conditional correlation: on properties and estimation. *Journal of Business & Economic Statistics* 31:282–299.
- Amihud, Y. 2002. Illiquidity and stock returns: cross-section and time-series effects. *Journal of Financial Markets* 5:31–56.
- Amihud, Y., A. Hameed, W. Kang, and H. Zhang. 2015. The illiquidity premium: International evidence. *Journal of Financial Economics* 117:350–368.
- Amihud, Y., and H. Mendelson. 1986. Asset pricing and bid-ask spread. *Journal of Financial Economics* 17:223–249.
- Amihud, Y., and H. Mendelson. 2015. The pricing of illiquidity as a characteristic and as risk. *Multinational Financial Journal* 19:149–168.
- Bekaert, G., and C. R. Harvey. 1995. Time-varying world market integration. *Journal of Finance* 40:403–444.
- Bekaert, G., C. R. Harvey, and C. Lundblad. 2007. Liquidity and expected returns: Lessons from emerging markets. *Review of Financial Studies* 20:1783–1831.
- Bekaert, G., C. R. Harvey, C. Lundblad, and S. Siegel. 2011. What segments equity markets? *Review of Financial Studies* 42:3841–3890.
- Bollerslev, T., R. Engle, and J. M. Wooldridge. 1988. A capital asset pricing model with time-varying covariances. *Journal of Political Economy* 96:116–131.
- Bollerslev, T., and J. M. Wooldridge. 1992. Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariance. *Economic Reviews* 11:143–172.
- Bongaerts, D., F. D. Jong, and J. Driessen. 2011. Derivative pricing with liquidity risk: theory and evidence from the credit default swap market. *Journal of Finance* 66:203–240.

- Brennan, M., T. Chordia, and A. Subrahmanyam. 1998. Alternative factor specifications, security characteristics and the cross-section of expected returns. *Journal of Financial Economics* 49:345–374.
- Campbell, J. Y., and L. M. Viceira. 2002. *Strategic asset allocation*. Oxford University Press.
- Carrieri, F., I. Chaieb, and V. Errunza. 2013. Do implicit barriers matter for globalization. *Review of Financial Studies* 26:1694–1739.
- Chaieb, I., and V. Errunza. 2007. International asset pricing under segmentation and PPP deviations. *Journal of Financial Economics* 86:543–578.
- Chordia, T., R. Roll, and A. Subrahmanyam. 2000. Commonality in liquidity. *Journal of Financial Economics* 56:3–28.
- Christoffersen, P., K. Jacobs, X. Jin, and H. Langlois. 2016. Dynamic dependence and diversification in corporate credit. *Working paper, HEC Paris* .
- Christoffersen, P., and H. Langlois. 2013. The joint dynamics of equity market factors. *The Journal of Financial and Quantitative Analysis* 48:1371–1404.
- Constantinides, G. 1986. Capital market equilibrium with transaction costs. *Journal of Political Economy* 94:842–862.
- Corwin, S. A., and P. Schultz. 2012. A simple way to estimate bid-ask spreads from daily high and low prices. *Journal of Finance* 2:719–759.
- de Jong, F. C. J. M., and F. A. Roon. 2005. Time-varying market integration and expected returns in emerging markets. *Journal of Financial Economics* 78:583–613.
- de Santis, G., and B. Gerard. 1997. Asset pricing and portfolio diversification with time-varying risk. *Journal of Finance* 52:1881–1912.
- Domowitz, I., J. Glen, and A. Madhavan. 2001. Liquidity, volatility and equity trading costs across countries and over time. *International Finance* 4:221–255.
- Dumas, B., and B. Solnik. 1995. The world price of foreign exchange rate risk. *Journal of Finance* 50:445–479.

- Errunza, V., and E. Losq. 1985. International asset pricing under mild segmentation: Theory and test. *Journal of Finance* 40:105–124.
- Errunza, V., and H. Ta. 2015. The impact of investability on asset valuation. *Journal of Financial and Quantitative Analysis* 50:1135–1163.
- Eun, C., and S. Janakiramanan. 1986. A model of international asset pricing with a constraint on the foreign equity ownership. *Journal of Finance* 41:897–914.
- Ferson, W., and C. R. Harvey. 1993. The risk and predictability of international equity returns. *Review of Financial Studies* 6:527–566.
- Ferson, W., and C. R. Harvey. 1994. Source of risk and expected returns in global equity markets. *Journal of Banking and Finance* 18:775–803.
- Fong, K., C. W. Holden, and C. A. Trzcinka. 2014. What are the best liquidity proxies for global Research? *Working paper, Indiana University* .
- Goyenko, R., and S. Sarkissian. 2014. Treasury bond illiquidity and global equity returns. *Journal of Financial and Quantitative Analysis* 49:1227–1253.
- Grauer, F. L., R. H. Litzenberger, and R. E. Stehle. 1976. Sharing rules and equilibrium in an international capital market under uncertainty. *Journal of Financial Economics* 3:233–256.
- Griffin, J. M., P. J. Kelly, and F. Nardari. 2010. Do market efficiency measures yield correct inferences? A comparison of developed and emerging markets. *Review of Financial Studies* 23:3225–3277.
- Hagstromer, B., B. Hansson, and B. Nilsson. 2011. Conditional asset pricing with liquidity risk. *Working paper, Stockholm University* .
- Harvey, C. R. 1991. The world price of covariance risk. *Journal of Finance* 46:111–157.
- Hasbrouck, J., and D. Seppi. 2001. Common factors in prices, order flows and liquidity. *Journal of Financial Economics* 59:383–411.
- Huang, J., and J. Wang. 2009. Liquidity and market crashes. *Review of Financial Studies* 22:2607–2643.

- Huberman, G., and D. Halka. 2001. Systematic liquidity. *Journal of Financial Research* 24:161–178.
- Ince, O. S., and R. B. Porter. 2006. Individual equity return data from Thomson Datastream: Handle with care! *Journal of Financial Research* 29:463–479.
- Johnson, T. 2006. Dynamic liquidity in endowment economies. *Journal of Financial Economics* 80:531–62.
- Karolyi, A., K.-H. Lee, and M. van Dijk. 2012. Understanding commonality in liquidity around the world. *Journal of Financial Economics* 105:82–112.
- Karolyi, A., and Y. Wu. 2012. The role of investability restrictions on size, value and momentum in international stock returns. *Working paper, Cornell University* .
- Kim, S.-H., and K.-H. Lee. 2014. Pricing of liquidity risks: evidence from multiple liquidity measures. *Journal of Empirical Finance* 25:112–133.
- Koch, A., S. Ruenzi, and L. Starks. 2014. Commonality in liquidity - A demand side explanation. *Working paper* .
- Korajczyk, R. A., and R. Sadka. 2008. Pricing the commonality across alternative measures of liquidity. *Journal of Financial Economics* 87:45–72.
- Lang, M., K. Lins, and M. Maffett. 2012. Transparency, liquidity, and valuation: International evidence. *Journal of Accounting Research* 50:729–779.
- Lee, K.-H. 2011. The world price of liquidity risk. *Journal of Financial Economics* 99:136–161.
- Lesmond, D. 2005. Liquidity of emerging markets. *Journal of Financial Economics* 77:411–452.
- Lesmond, D., J. Ogden, and C. Trzcinka. 1999. A new estimate of transaction costs. *Review of Financial Studies* 12:1113–1141.
- Lesmond, D., M. Schill, and C. Zhou. 2004. The illusory nature of momentum profits. *Journal of Financial Economics* 71:349–380.
- Levine, R., and S. L. Schmukler. 2006. Internationalization and stock market liquidity. *Review of Finance* 10:153–187.

- Mancini, L., A. Ranaldo, and J. Wrampelmeyer. 2013. Liquidity in the foreign exchange market: measurement, commonality, and risk premiums. *Journal of Finance* 68:1805–1841.
- Mei, J., J. Scheinkman, and W. Xiong. 2009. Speculative trading and stock prices: Evidence from Chinese A-B share premia. *Annals of Economics and Finance* 10:225–255.
- Pastor, L., and R. F. Stambaugh. 2003. Liquidity risk and expected stock returns. *Journal of Political Economy* 111:642–685.
- Sadka, R. 2006. Momentum and post-earnings announcement drift anomalies: The role of liquidity risk. *Journal of Financial Economics* 80:309–350.
- Shanno, D. F. 1985. On Broyden-Fletcher-Goldfarb-Shanno method. *Journal of Optimization Theory and Applications* 46:87–94.
- Stulz, R. 1981. On the effects of barriers to international investment. *Journal of Finance* 36:923–934.
- Watanabe, A., and M. Watanabe. 2008. Time-varying liquidity risk and the cross section of stock returns. *Review of Financial Studies* 21:2449–2486.

Figures and Tables

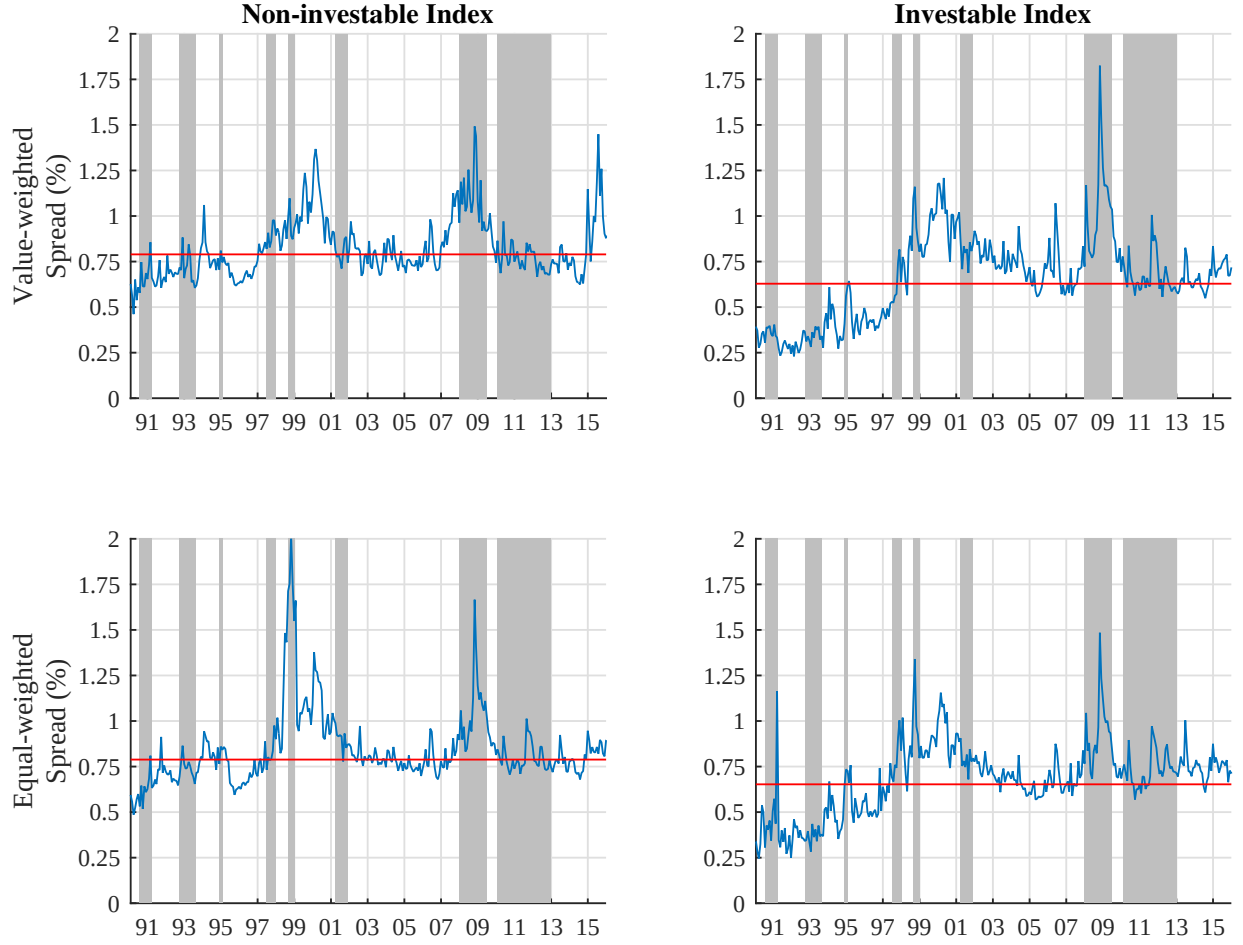


Figure 1 Bid-ask spread for investable and non-investable securities averaged across emerging markets

We report the monthly bid-ask spreads averaged across all 21 emerging markets. For each country, we build value-weighted indices that contain either non-investable or investable stocks. Then, we build cross-country averages of these portfolios' bid-ask spreads. The top row reports on the market capitalization weighted country average, and the bottom row shows equal-weighted country averages. The left column reports the bid-ask spread measures for portfolios of non-investable stocks and the right column for portfolios of investable stocks. In each graph, the horizontal line reports the time-series average of the conditional bid-ask spread measure. We use the high and low price based bid-ask spread measure of [Corwin and Schultz \(2012\)](#). The sample period is monthly from January 1990 to December 2015. Gray areas indicate NBER recessions (July 1990 to March 1991, March to November 2001, and December 2007 to June 2009), the European exchange rate mechanism crisis (September 1992 to August 1993), the Tequila crisis (December 1994 to January 1995), the East Asia crisis (June to December 1997), the Russian default and Long-Term Capital Management crisis (August to December 1998), and the Euro-sovereign debt crisis (January 2010 to December 2012).

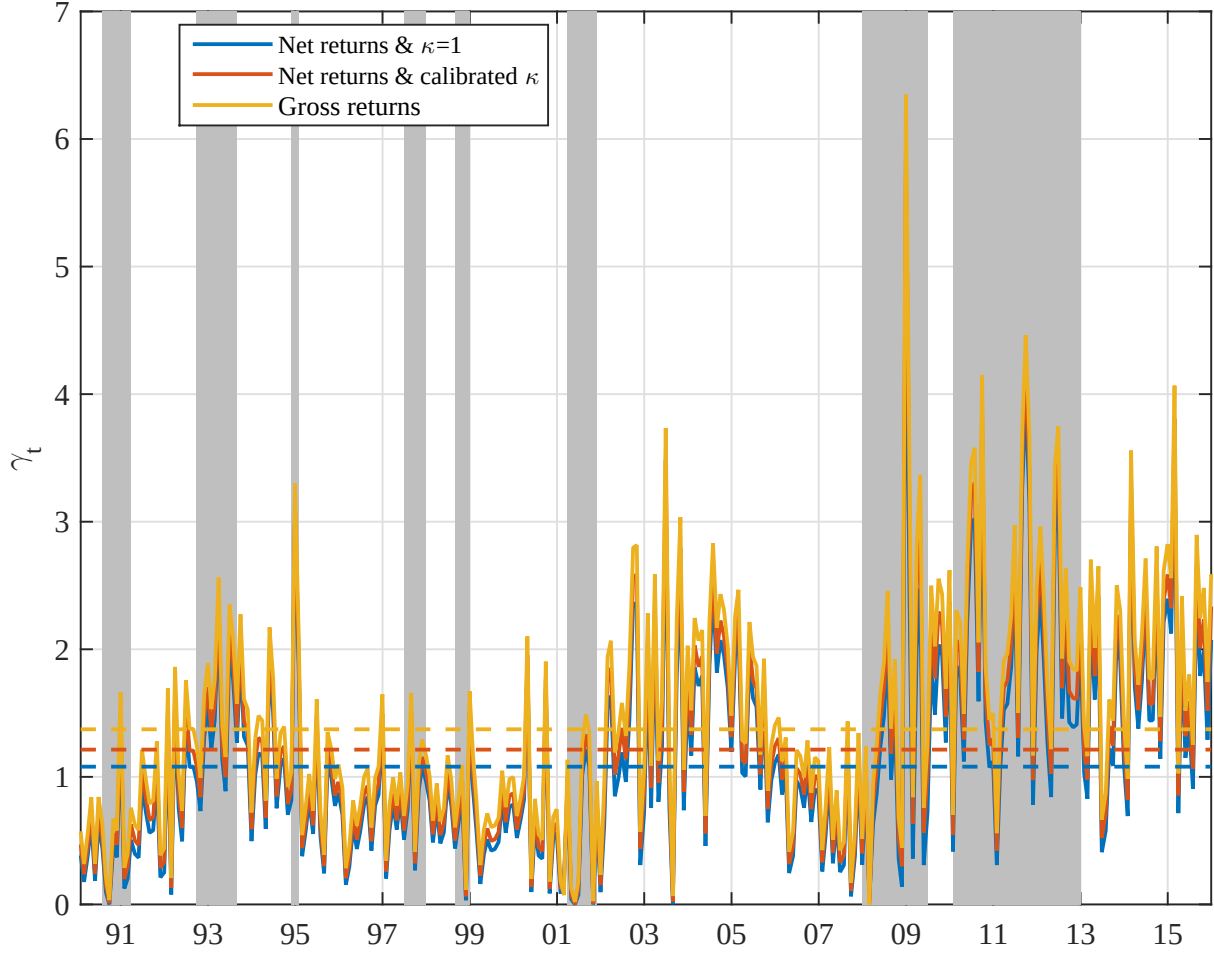


Figure 2 Price of World Covariance Risk γ_t

We report the conditional price of world covariance risk, γ_t , estimated with three specifications: on net returns with $\kappa_W = 1$, on net returns with κ_W calibrated to the average turnover, and on gross returns. The horizontal dashed lines reports the time-series averages of the conditional prices. The prices of risk are estimated by the methodology of Section 3.2. The sample period is monthly from January 1990 to December 2015. Gray areas indicate NBER recessions (July 1990 to March 1991, March to November 2001, and December 2007 to June 2009), the European exchange rate mechanism crisis (September 1992 to August 1993), the Tequila crisis (December 1994 to January 1995), the East Asia crisis (June to December 1997), the Russian default and Long-Term Capital Management crisis (August to December 1998), and the Euro-sovereign debt crisis (January 2010 to December 2012).

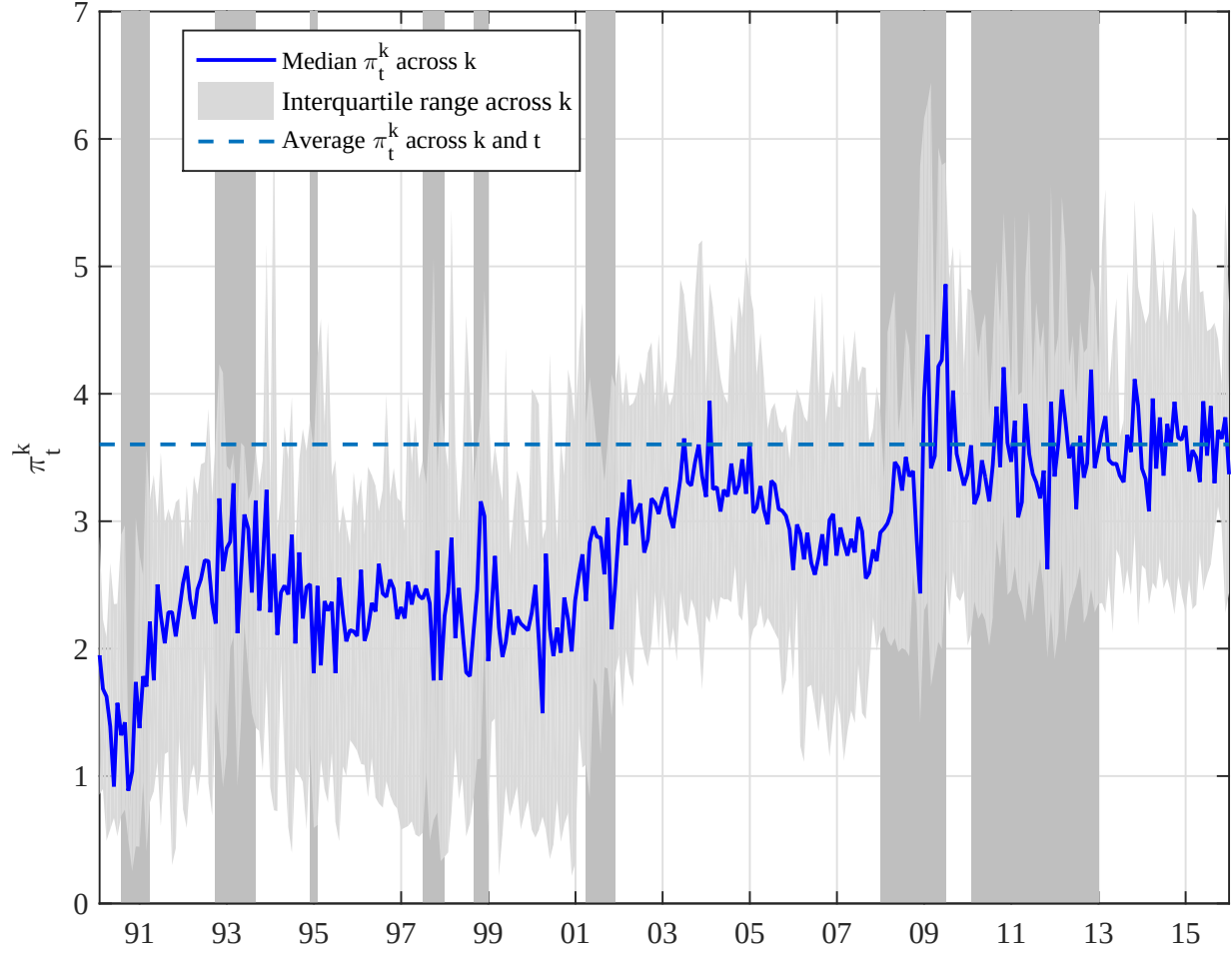


Figure 3 Price of Local Covariance Risk π_t^k

We report the time-varying median price of conditional covariance risk, π_t^k , estimated on net returns with our main specification. The light gray area reports the 25th to 75th range of prices computed across all available countries at each point in time. The horizontal dashed line is the average price across countries and over time. The prices of risk are estimated by the methodology of Section 3.2 applied to net returns. See Equation system 11. The sample period is monthly from January 1990 to December 2015. Gray areas indicate NBER recessions (July 1990 to March 1991, March to November 2001, and December 2007 to June 2009), the European exchange rate mechanism crisis (September 1992 to August 1993), the Tequila crisis (December 1994 to January 1995), the East Asia crisis (June to December 1997), the Russian default and Long-Term Capital Management crisis (August to December 1998), and the Euro-sovereign debt crisis (January 2010 to December 2012).

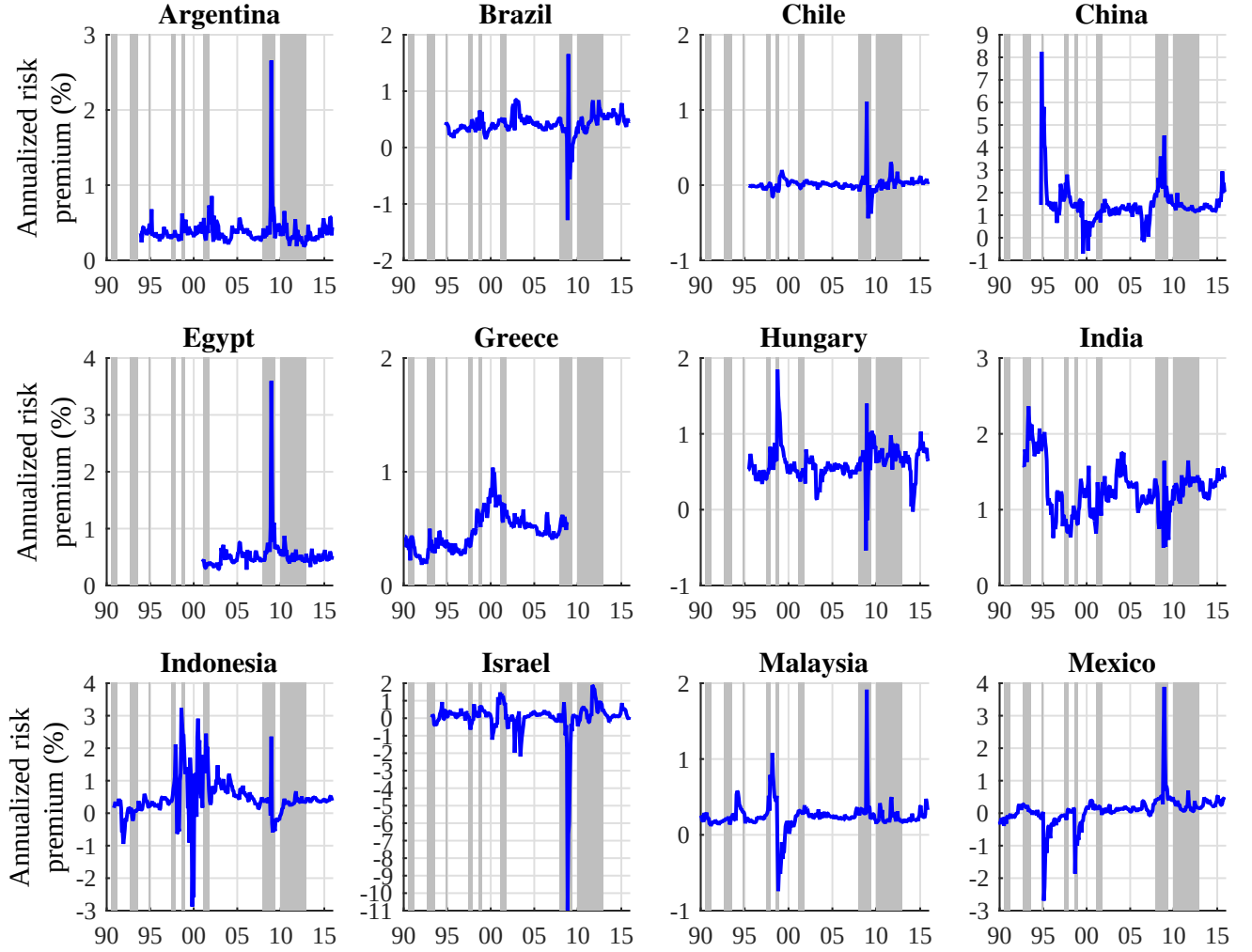


Figure 4 Illiquidity Level and Risk Premiums

For each EM, we report the annualized total risk premium coming from illiquidity terms,

$$\begin{aligned} \kappa_{N^k} E_t [c_{N^k,t+1}] &+ \gamma_{t+1} (cov_t(c_{N^k,t+1}, c_{W,t+1}) - cov_t(r_{N^k,t+1}, c_{W,t+1}) - cov_t(r_{N^k,t+1}, r_{W,t+1})) \\ &+ \pi_{t+1}^k (var_t(c_{N^k,t+1}) - 2cov_t(r_{N^k,t+1}, c_{N^k,t+1}) + cov_t(r_{N^k,t+1}, c_{DP,t+1}) \\ &\quad + cov_t(c_{N^k,t+1}, r_{DP,t+1}) - cov_t(c_{N^k,t+1}, c_{DP,t+1})) \end{aligned}$$

κ_{N^k} is calibrated using the value-weighted average of monthly turnover of all stocks in the non-investable market portfolio. The sample period varies across countries depending on data availability. Gray areas indicate NBER recessions (July 1990 to March 1991, March to November 2001, and December 2007 to June 2009), the European exchange rate mechanism crisis (September 1992 to August 1993), the Tequila crisis (December 1994 to January 1995), the East Asia crisis (June to December 1997), the Russian default and Long-Term Capital Management crisis (August to December 1998), and the Euro-sovereign debt crisis (January 2010 to December 2012).

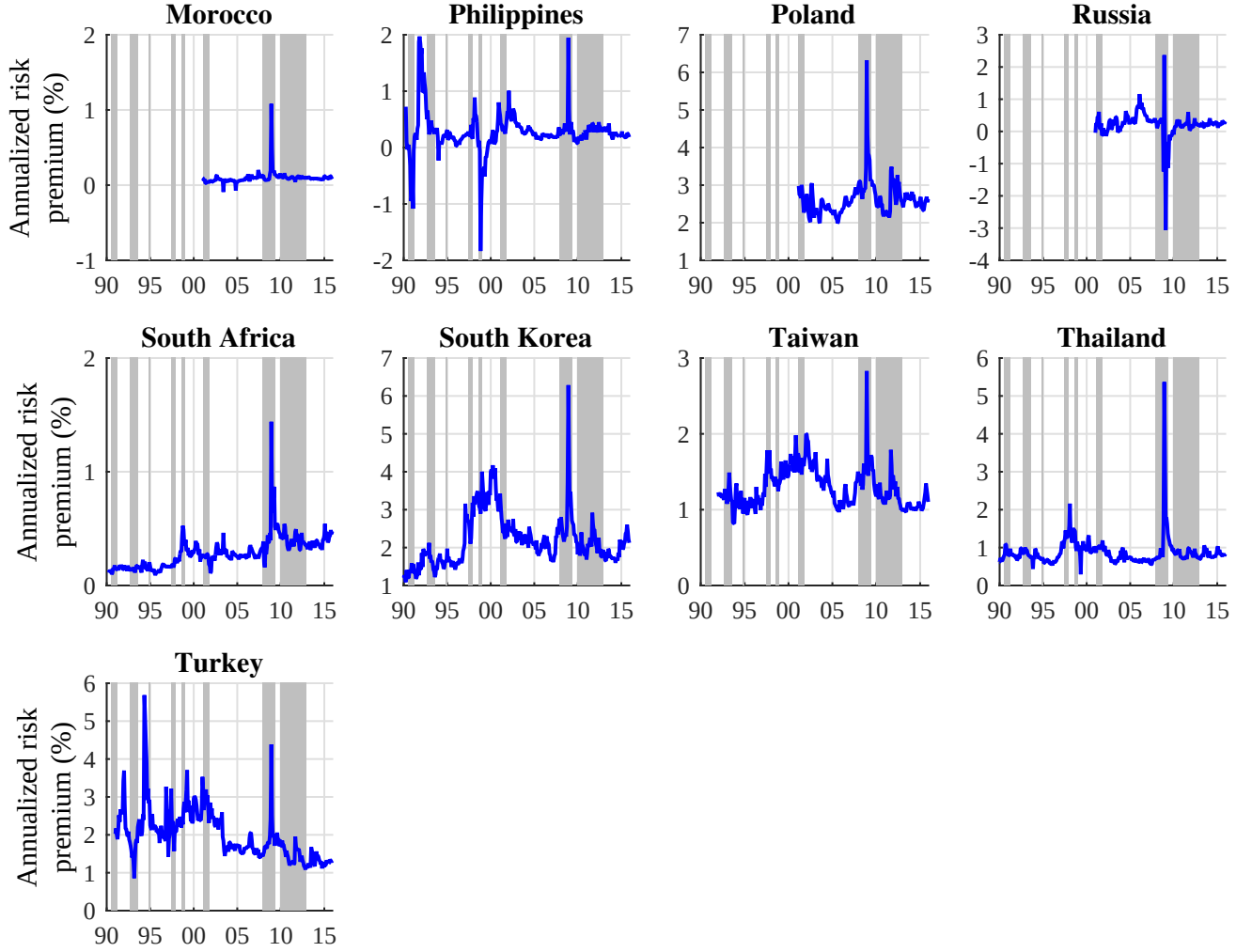


Figure 4 Illiquidity Level and Risk Premiums (Continued)

For each EM, we report the annualized total risk premium coming from illiquidity terms,

$$\begin{aligned} \kappa_{N^k} E_t [c_{N^k,t+1}] + \gamma_{t+1} (& cov_t(c_{N^k,t+1}, c_{W,t+1}) - cov_t(r_{N^k,t+1}, c_{W,t+1}) - cov_t(r_{N^k,t+1}, r_{W,t+1})) \\ & + \pi_{t+1}^k (var_t(c_{N^k,t+1}) - 2cov_t(r_{N^k,t+1}, c_{N^k,t+1}) + cov_t(r_{N^k,t+1}, c_{DP,t+1}) \\ & + cov_t(c_{N^k,t+1}, r_{DP,t+1}) - cov_t(c_{N^k,t+1}, c_{DP,t+1})) \end{aligned}$$

κ_{N^k} is calibrated using the value-weighted average of monthly turnover of all stocks in the non-investable market portfolio. The sample period varies across countries depending on data availability. Gray areas indicate NBER recessions (July 1990 to March 1991, March to November 2001, and December 2007 to June 2009), the European exchange rate mechanism crisis (September 1992 to August 1993), the Tequila crisis (December 1994 to January 1995), the East Asia crisis (June to December 1997), the Russian default and Long-Term Capital Management crisis (August to December 1998), and the Euro-sovereign debt crisis (January 2010 to December 2012).

Table 1 - Number of stocks and summary statistics of returns and liquidity measure by country and year.

For each country, we report the total number of local stocks over the whole period (column (1)), the number of local stocks at the start of the sample and every five years (columns (2)-(9)), the average and volatility of the returns and bid-ask spread (columns (10)-(13)). We report each statistic for the portfolio of local stocks, of local non-investable stocks, and local investable stocks. For each country, we report in italics in column (1) the proportion of investable stocks and in column (11) the ratio of the average bid-ask spreads of the non-investable portfolio to the investable portfolio. The average returns and return volatilities are annualized. All portfolios are constructed from individual stocks as detailed in the Data section of the paper. Returns are denominated in USD and in excess of the one-month US T-bill rate. The bid-ask spread is the Corwin and Schultz (2012) approximation using high and low prices. The averages and volatility are computed over the sample period when both the non-investable and the investable portfolios are available.

Country	Number of stocks in the sample									Average (%)		Volatility (%)	
	N (1)	1988 (2)	1993 (3)	1998 (4)	2003 (5)	2008 (6)	2013 (7)	2014 (8)	2015 (9)	Return (10)	Bid-ask spread (11)	Return (12)	Bid-ask spread (13)
Argentina	<i>17.58%</i>										<i>1.15</i>		
All	91	0	38	26	37	28	39	31	36	3.60	0.61	37.64	0.26
Non-Investable	88	0	36	21	31	19	28	20	25	1.90	0.66	40.54	0.24
Investable	16	0	2	5	6	9	11	11	11	4.99	0.57	39.03	0.30
Brazil	<i>7.69%</i>										<i>1.22</i>		
All	273	0	0	22	42	117	156	150	146	7.97	0.85	38.25	0.20
Non-Investable	265	0	0	21	33	104	139	133	129	8.47	0.94	40.05	0.21
Investable	21	0	0	1	9	13	17	17	17	6.56	0.77	43.01	0.29
Chile	<i>9.93%</i>										<i>0.70</i>		
All	141	0	5	23	33	40	45	43	35	2.84	0.47	23.46	0.16
Non-Investable	133	0	5	17	24	31	38	37	29	3.52	0.41	24.89	0.14
Investable	14	0	0	6	9	9	7	6	6	2.47	0.59	24.01	0.28
China	<i>7.60%</i>										<i>1.03</i>		
All	3013	0	87	723	1260	1562	2336	2162	2218	8.04	0.90	28.99	0.24
Non-Investable	2832	0	86	708	1117	1377	2140	1974	2035	8.84	0.91	29.86	0.25
Investable	229	0	1	15	143	185	196	188	183	2.23	0.89	37.67	0.31
Egypt	<i>6.17%</i>										<i>1.19</i>		
All	162	0	0	44	45	87	97	96	74	7.13	0.79	29.32	0.25
Non-Investable	156	0	0	41	40	82	94	93	70	3.58	0.84	28.75	0.28
Investable	10	0	0	3	5	5	3	3	4	14.12	0.71	36.55	0.27
Greece	<i>8.61%</i>										<i>1.47</i>		
All	418	16	83	183	302	204	100	82	62	1.87	0.82	33.94	0.23
Non-Investable	415	16	83	170	282	181	80	63	43	0.71	1.00	38.33	0.23
Investable	36	0	0	13	20	23	20	19	19	3.48	0.68	33.10	0.22

Country	Number of stocks in the sample									Average (%)		Volatility (%)	
	N	1988	1993	1998	2003	2008	2013	2014	2015	Return	Bid-ask spread	Return	Bid-ask spread
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Hungary	42.42%										1.19		
All	66	0	3	22	27	22	22	19	19	6.65	0.66	35.29	0.24
Non-Investable	52	0	3	7	9	11	9	8	10	-2.36	0.79	38.61	0.32
Investable	28	0	0	15	18	11	13	11	9	6.80	0.66	35.98	0.24
India	0.93%										0.82		
All	2903	0	183	148	529	799	489	921	756	3.45	1.32	34.39	0.26
Non-Investable	2903	0	183	146	528	793	480	917	755	3.38	1.32	34.70	0.26
Investable	27	0	0	2	1	6	9	4	1	-14.75	1.62	65.13	0.83
Indonesia	16.00%										1.29		
All	500	0	39	85	95	90	234	256	214	-0.63	1.21	44.68	0.84
Non-Investable	480	0	39	83	73	40	174	190	157	-3.18	1.39	45.71	1.01
Investable	80	0	0	2	22	50	60	66	57	1.36	1.07	45.18	0.72
Israel	7.72%										1.59		
All	557	0	27	85	104	154	198	207	177	6.75	0.51	23.42	0.18
Non-Investable	532	0	26	81	91	135	178	186	156	4.34	0.60	27.04	0.23
Investable	43	0	1	4	13	19	20	21	21	20.59	0.38	27.09	0.15
Malaysia	1.20%										1.00		
All	998	102	295	379	575	271	458	500	464	6.40	0.62	30.77	0.22
Non-Investable	998	96	286	372	571	270	457	499	463	6.40	0.62	31.08	0.22
Investable	12	6	9	7	4	1	1	1	1	6.20	0.62	31.78	0.27
Mexico	26.51%										1.03		
All	215	13	53	46	41	49	65	68	77	8.50	0.58	30.80	0.23
Non-Investable	196	12	50	37	29	10	27	29	38	9.90	0.58	31.74	0.22
Investable	57	1	3	9	12	39	38	39	39	5.17	0.56	32.50	0.24
Morocco	5.19%										1.35		
All	77	0	0	9	11	42	15	19	14	7.23	0.44	17.57	0.13
Non-Investable	76	0	0	8	9	39	13	17	13	8.46	0.51	19.36	0.19
Investable	4	0	0	1	2	3	2	2	1	6.30	0.38	18.16	0.13
Philippines	12.24%										1.63		
All	245	1	43	46	24	52	91	131	118	4.23	0.69	30.37	0.16
Non-Investable	218	1	33	41	20	48	89	127	117	2.83	0.76	31.66	0.20
Investable	30	0	10	5	4	4	2	4	1	6.78	0.46	31.64	0.25

Country	Number of stocks in the sample									Average (%)		Volatility (%)	
	N	1988	1993	1998	2003	2008	2013	2014	2015	Return	Bid-ask spread	Return	Bid-ask spread
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Poland	3.32%										1.08		
All	843	0	0	2	62	277	474	450	452	7.91	0.77	31.04	0.19
Non-Investable	835	0	0	1	61	268	457	430	429	8.28	0.80	31.05	0.18
Investable	28	0	0	1	1	9	17	20	23	8.52	0.74	32.35	0.25
Russia	20.89%										1.36		
All	292	0	0	7	19	98	116	106	122	11.72	0.84	35.15	0.36
Non-Investable	261	0	0	3	5	59	81	73	90	17.13	1.11	39.46	0.46
Investable	61	0	0	4	14	39	35	33	32	10.81	0.82	35.28	0.36
South Africa	15.05%										0.98		
All	731	0	35	254	142	162	183	191	181	3.18	0.64	26.59	0.29
Non-Investable	684	0	12	226	117	107	131	142	134	5.02	0.63	26.65	0.27
Investable	110	0	23	28	25	55	52	49	47	3.59	0.64	27.69	0.31
South Korea	1.04%										1.40		
All	2801	375	630	709	1480	1699	1673	1714	1800	-0.94	0.98	37.46	0.30
Non-Investable	2798	375	630	703	1466	1681	1659	1700	1786	-3.20	1.10	40.34	0.32
Investable	29	0	0	6	14	18	14	14	14	4.19	0.78	35.55	0.28
Taiwan	5.83%										1.13		
All	1029	0	236	444	668	703	819	825	824	-0.17	0.70	27.06	0.18
Non-Investable	1029	0	236	432	628	680	786	794	793	-0.39	0.73	27.26	0.18
Investable	60	0	0	12	40	23	33	31	31	0.22	0.64	30.04	0.19
Thailand	7.23%										1.11		
All	802	66	259	110	250	266	429	486	504	2.49	0.81	36.99	0.26
Non-Investable	787	66	259	86	216	225	385	442	461	1.77	0.87	37.16	0.27
Investable	58	0	0	24	34	41	44	44	43	2.92	0.78	37.74	0.26
Turkey	8.50%										1.04		
All	494	42	136	261	276	298	373	368	367	8.13	1.20	48.71	0.49
Non-Investable	493	42	136	253	246	275	366	362	364	9.16	1.22	48.19	0.48
Investable	42	0	0	8	30	23	7	6	3	-0.85	1.17	51.51	0.57
All Countries		615	2152	3628	6022	7020	8412	8825	8660				
Proportion of investables		1.1%	2.3%	4.7%	7.1%	8.3%	7.1%	6.7%	6.5%				

Table 2: Pricing tests of the model

We report estimation results and test statistics for the asset pricing model in Equation (11). Panel A reports the specification tests of the world covariance risk. Column (1) reports the starting date of the estimation of world market equation. Columns (2)-(7) report Wald test statistics and their p -values for the test hypotheses of the three model specifications; (1) the main specification in which we use covariances of net returns and we impose the model-implied restriction $\kappa = 1$, (2) the alternative specification in which we use covariances of net returns and calibrate κ to the average monthly turnover, and (3) the benchmark specification in which we use covariance of gross returns and set κ to zero. Columns (8)-(10) report the pricing error of the world market equation from the three model specifications. Panel B reports the conditional local price of risk tests from the main and benchmark specifications. Column (1) reports the starting date of the model estimation for each country. The next four columns (2)-(5) report Wald tests and below their p -values for the test hypotheses. Columns (6)-(9) report the intercept estimates and t -ratios of non-investable market portfolios and their diversification portfolios from the two model specifications. Significant values at the 5% level are bolded.

Panel A. Specification tests of world market covariance risk

Null hypothesis	Start date of estimation	for insignificant world market risk ($\lambda_{j,W}=0$ for all j) (d.f. = 4)			for constant world market risk ($\lambda_{j,W}=0$ for $j > 1$) (d.f. = 3)			Intercept		
		net returns & $\kappa = 1$	net returns & calibrated κ	gross returns	net returns & $\kappa = 1$	net returns & calibrated κ	gross returns	net returns & $\kappa = 1$	net returns & calibrated κ	gross returns
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
World market	April 1984	1869.20 0.00	2310.45 0.00	2493.09 0.00	186.02 0.00	219.48 0.00	231.01 0.00	0.06% 0.08	0.64% 0.91	0.66% 0.88

Panel B. Specification tests of conditional local market risk

Null hypothesis	Start date of estimation	for insignificant local market risk ($\lambda_{i,k}=0$ for all i) (d.f. = 4)		for constant local market risk ($\lambda_{i,k}=0$ for $i > 1$) (d.f. = 3)		Intercept			
		net returns & $\kappa = 1$	gross returns	net returns & $\kappa = 1$	gross returns	net returns & $\kappa = 1$		gross returns	
	(1)	(2)	(3)	(4)	(5)	Non-investable portfolio	Diversification portfolio	Non-investable portfolio	Diversification portfolio
Argentina	November 1993	2566.25 0.00	829.97 0.00	245.44 0.00	130.76 0.00	-1.07% 0.35	0.73% 1.30	1.20% 0.36	1.98% 3.62
Brazil	November 1994	778.27 0.00	7293.39 0.00	81.37 0.00	396.65 0.00	0.38% 0.24	0.56% 1.10	0.69% 1.12	0.97% 1.91
Chile	July 1995	1148.61 0.00	665.04 0.00	104.77 0.00	57.36 0.00	-0.19% 0.17	0.09% 0.30	-0.24% 0.18	0.21% 0.74
China	October 1994	7241.77 0.00	8256.33 0.00	463.37 0.00	495.70 0.00	-1.09% 0.82	0.05% 0.19	-0.68% 0.42	0.37% 1.38
Egypt	January 2001	1436.67 0.00	435.37 0.00	80.26 0.00	36.26 0.00	-1.17% 0.40	0.64% 1.34	-0.46% 0.21	1.08% 2.22
Greece	March 1990	140.40 0.00	155.51 0.00	6.33 0.10	6.66 0.08	0.05% 0.06	0.05% 0.32	0.67% 0.84	0.42% 2.77
Hungary	June 1995	741.00 0.00	1085.14 0.00	120.08 0.00	427.84 0.00	-0.80% 0.40	0.51% 1.28	0.02% 0.03	1.17% 2.99
India	October 1992	2274.12 0.00	1472.30 0.00	340.72 0.00	248.01 0.00	-2.89% 0.89	0.24% 0.61	-1.62% 0.50	0.68% 1.73
Indonesia	November 1990	20295.83 0.00	19568.68 0.00	973.43 0.00	1391.73 0.00	-1.14% 1.23	0.10% 0.21	-0.23% 0.33	0.76% 1.78
Israel	April 1993	3789.48 0.00	2708.51 0.00	704.11 0.00	394.29 0.00	-2.54% 1.21	0.26% 0.86	-2.32% 0.97	0.68% 2.25
Malaysia	May 1986	3241.25 0.00	1528.86 0.00	1583.42 0.00	31.56 0.00	-0.12% 0.34	-0.04% 0.18	0.64% 2.03	0.60% 2.32
Mexico	July 1989	739.96 0.00	1037.35 0.00	59.81 0.00	73.26 0.00	0.56% 0.64	0.41% 0.80	1.25% 1.41	1.21% 2.45
Morocco	January 2001	169.94 0.00	265.04 0.00	26.10 0.00	13.13 0.00	-0.30% 0.11	-0.08% 0.23	0.39% 0.20	0.22% 0.59
Philippines	April 1990	817.26 0.00	1905.71 0.00	93.99 0.00	137.81 0.00	-0.59% 0.26	0.21% 0.57	0.20% 0.17	0.72% 1.90
Poland	March 2001	1744.33 0.00	477.21 0.00	142.25 0.00	21.19 0.00	0.09% 0.14	0.26% 0.61	0.57% 0.20	1.04% 2.50

Null hypothesis	Start date of estimation	for insignificant local market risk ($\lambda_{i,k}=0$ for all i) (d.f. = 4)		for constant local market risk ($\lambda_{i,k}=0$ for $i > 1$) (d.f. = 3)		Intercept			
		net returns & $\kappa = 1$	gross returns	net returns & $\kappa = 1$	gross returns	net returns & $\kappa = 1$		gross returns	
						Non-investable portfolio	Diversification portfolio	Non-investable portfolio	Diversification portfolio
Russia	January 2001	624.25 0.00	446.73 0.00	53.20 0.00	32.92 0.00	-0.19% 0.07	0.45% 0.66	0.58% 0.20	0.96% 1.48
South Africa	April 1990	1772.19 0.00	1026.74 0.00	145.12 0.00	150.58 0.00	0.05% 0.06	0.29% 1.05	0.78% 1.37	0.84% 3.03
South Korea	November 1987	12680.56 0.00	8.62 0.03	1251.83 0.00	0.77 0.68	-1.03% 2.08	0.28% 0.90	0.78% 1.61	0.74% 2.48
Taiwan	January 1992	1908.85 0.00	1732.38 0.00	431.07 0.00	342.66 0.00	-0.56% 0.98	-0.16% 0.53	0.06% 0.11	0.32% 1.06
Thailand	May 1987	6359.47 0.00	4173.62 0.00	584.35 0.00	527.96 0.00	-0.25% 0.27	0.28% 0.84	0.50% 0.55	0.94% 2.79
Turkey	January 1991	59088.60 0.00	6416.39 0.00	5756.52 0.00	1243.05 0.00	-1.22% 1.34	0.11% 0.13	0.09% 0.09	0.75% 1.01

Table 3: Illiquidity Level and Risk Premiums

For each EM, we report the contribution to expected returns of all terms that involve illiquidity. We report in column (1) the average turnover which is computed as the value-weighted average of monthly turnover of all stocks in the non-investable market portfolio when data is available. Columns (2) to (4) contain the time-series average, minimum, and maximum of κ times the expected transaction cost. Columns (5) to (7) contain the time-series average, minimum, and maximum of the illiquidity component of the world risk premium. Columns (8) to (10) contain the time-series average, minimum, and maximum of the illiquidity component of the conditional local risk premium. Columns (11) to (13) contain the time-series average, minimum, and maximum of the sum of all illiquidity components. All figures are in percentage and annualized.

	Average turnover	Illiquidity level			Sum of world illiquidity risk premiums			Sum of conditional local illiquidity risk premiums			Sum of illiquidity premiums		
		Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Argentina	1.662	0.127	0.122	0.225	0.073	-0.001	2.163	0.181	-0.012	0.696	0.382	0.181	2.658
Brazil	3.507	0.388	0.381	0.793	0.080	-0.011	2.613	-0.051	-2.280	0.463	0.418	-1.286	1.656
Chile	1.214	0.058	0.058	0.098	0.042	-0.006	1.690	-0.087	-0.669	0.121	0.014	-0.441	1.111
China	14.513	1.541	1.459	2.607	0.027	-0.142	2.261	-0.099	-3.238	6.848	1.469	-0.706	8.247
Egypt	3.854	0.381	0.365	0.814	0.106	0.000	2.522	0.047	-0.298	0.365	0.534	0.278	3.597
Greece	4.587	0.451	0.442	0.839	0.023	-0.050	0.215	0.000	-0.089	0.099	0.474	0.182	1.036
Hungary	4.561	0.413	0.398	0.887	0.112	0.000	2.851	0.084	-2.187	0.814	0.609	-0.539	1.849
India	6.418	1.108	1.086	2.016	0.059	0.000	1.631	0.100	-1.464	0.625	1.266	0.497	2.363
Indonesia	3.059	0.437	0.336	2.118	0.073	-0.026	3.110	-0.045	-4.480	2.571	0.466	-2.876	3.237
Israel	3.503	0.243	0.246	0.616	0.068	-0.025	3.099	-0.261	-11.808	1.247	0.050	-10.982	1.877
Malaysia	2.736	0.203	0.195	0.547	0.057	-0.075	1.679	-0.024	-1.229	0.555	0.236	-0.742	1.915
Mexico	3.591	0.245	0.234	0.526	0.070	0.000	2.848	-0.228	-3.060	1.530	0.086	-2.693	3.887
Morocco	1.335	0.081	0.081	0.128	0.012	-0.208	0.931	0.002	-0.111	0.094	0.096	-0.091	1.081
Philippines	1.551	0.141	0.142	0.331	0.040	-0.052	1.772	0.090	-1.997	1.659	0.271	-1.835	1.970
Poland	25.746	2.458	2.412	3.391	0.122	0.000	3.025	0.040	-0.161	0.493	2.620	1.989	6.324
Russia	1.037	0.132	0.119	0.383	0.126	-0.005	3.791	-0.034	-3.446	0.959	0.224	-3.057	2.369
South Africa	3.161	0.232	0.231	0.561	0.050	0.000	1.248	0.005	-0.338	0.159	0.287	0.091	1.440
South Korea	17.745	2.059	1.940	4.119	0.059	-0.097	2.708	0.000	-0.042	0.007	2.118	1.024	6.286
Taiwan	15.048	1.291	1.223	1.981	0.037	-0.019	1.273	-0.035	-0.504	0.069	1.294	0.817	2.829
Thailand	7.847	0.761	0.700	1.613	0.083	-0.071	3.289	0.024	-0.805	1.292	0.868	0.294	5.367
Turkey	12.659	1.887	1.818	4.322	0.053	-0.149	2.758	0.057	-1.045	3.122	1.997	0.857	5.685

Table 4: Alternative Choice of Non-Investable Securities

We report estimation results and test statistics for the asset pricing model in Equation (11). We include all local stocks in the market portfolio of non-investable securities. We report the conditional local price of risk tests for the main specification, in which we use covariances of net returns and we impose the model-implied restriction $\kappa = 1$, and the benchmark specification in which we use covariance of gross returns and set κ to zero. Column (1) reports the starting date of the model estimation for each country. The next four columns (2)-(5) report Wald tests and below their p -values for the test hypotheses. Columns (6)-(9) report the intercept estimates and t -ratios of non-investable market portfolios and their diversification portfolios from the two model specifications. Significant values at the 5% level are bolded.

Null hypothesis	Start date of estimation	for insignificant local market risk ($\lambda_{i,k}=0$ for all i) (d.f. = 4)		for constant local market risk ($\lambda_{i,k}=0$ for $i > 1$) (d.f. = 3)		Intercept			
		net returns & $\kappa = 1$	gross returns	net returns & $\kappa = 1$	gross returns	net returns & $\kappa = 1$		gross returns	
						Non-investable portfolio	Diversification portfolio	Non-investable portfolio	Diversification portfolio
Argentina	November 1993	1834.07 0.00	515.08 0.00	483.70 0.00	36.28 0.00	0.46% 0.67	0.68% 1.27	1.02% 0.72	1.41% 2.33
Brazil	November 1994	1025.73 0.00	2708.39 0.00	15.15 0.00	82.27 0.00	0.18% 0.38	0.31% 0.65	0.86% 1.81	0.76% 1.63
Chile	July 1995	177.25 0.00	246.33 0.00	15.44 0.00	26.85 0.00	0.39% 0.43	0.37% 1.16	0.40% 0.35	0.60% 1.89
China	October 1994	6634.48 0.00	6908.16 0.00	329.18 0.00	399.31 0.00	-2.66% 1.07	0.14% 0.48	-2.13% 0.85	0.64% 2.24
Egypt	January 2001	583.19 0.00	676.80 0.00	47.67 0.00	54.88 0.00	-0.83% 0.29	0.47% 1.10	0.08% 0.03	0.95% 2.16
Greece	March 1990	277.19 0.00	274.26 0.00	6.02 0.11	8.20 0.04	-0.11% 0.22	-0.09% 0.67	0.38% 0.75	0.13% 1.04
Hungary	June 1995	1363.47 0.00	1566.33 0.00	127.79 0.00	142.96 0.00	0.45% 0.85	1.00% 1.99	0.91% 1.46	1.44% 2.93
India	October 1992	2345.82 0.00	3764.79 0.00	245.50 0.00	376.78 0.00	-2.58% 0.77	0.42% 1.11	-2.07% 0.53	0.90% 2.40
Indonesia	November 1990	12193.04 0.00	8834.45 0.00	9421.26 0.00	7647.20 0.00	0.28% 0.52	0.52% 1.06	1.04% 1.99	1.13% 2.39
Israel	April 1993	1056.93 0.00	581.49 0.00	77.15 0.00	378.61 0.00	0.33% 0.48	0.36% 1.21	0.92% 2.50	0.89% 3.07
Malaysia	May 1986	4749.96 0.00	7142.13 0.00	762.41 0.00	1806.48 0.00	-0.08% 0.20	0.11% 0.45	0.32% 0.86	0.55% 2.17
Mexico	July 1989	2923.57 0.00	869.22 0.00	180.16 0.00	61.79 0.00	0.22% 0.34	0.60% 1.02	1.16% 1.76	1.26% 2.13
Morocco	January 2001	375.95 0.00	271.40 0.00	27.64 0.00	24.51 0.00	-0.07% 0.07	0.08% 0.69	0.31% 0.22	0.33% 2.45
Philippines	April 1990	3409.69 0.00	945.77 0.00	131.54 0.00	81.99 0.00	-0.34% 0.29	0.07% 0.21	0.58% 0.39	0.63% 1.88
Poland	March 2001	740.55 0.00	402.64 0.00	62.67 0.00	48.86 0.00	-0.35% 0.25	0.44% 1.03	0.47% 0.21	0.98% 2.36
Russia	January 2001	1132.42 0.00	283.36 0.00	83.50 0.00	219.28 0.00	0.53% 0.70	0.14% 0.19	1.17% 1.54	0.76% 1.04
South Africa	April 1990	1452.67 0.00	3959.84 0.00	202.96 0.00	599.97 0.00	-0.13% 0.19	0.18% 0.60	0.21% 0.38	0.65% 2.25
South Korea	November 1987	12792.60 0.00	10068.84 0.00	481.55 0.00	387.84 0.00	-0.22% 0.64	0.18% 0.61	0.63% 1.85	0.54% 1.84
Taiwan	January 1992	2626.89 0.00	2147.99 0.00	500.49 0.00	730.34 0.00	-0.63% 1.11	0.15% 0.78	-0.20% 0.26	0.63% 3.18
Thailand	May 1987	691.17 0.00	690.15 0.00	64.35 0.00	72.40 0.00	0.21% 0.54	0.32% 0.99	0.82% 2.18	0.87% 2.72
Turkey	January 1991	23919.76 0.00	14000.17 0.00	2320.67 0.00	2455.06 0.00	-1.81% 1.58	-0.26% 0.38	-0.77% 0.62	0.60% 0.83